



**MAGNETICUM**



# Numerical Simulations of physical processes driving galaxy evolution

## Lecture 2: Subgrid Physics

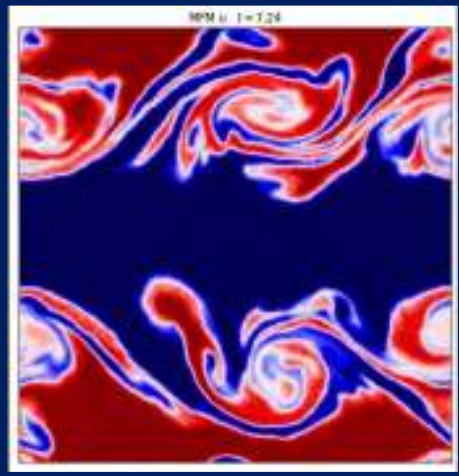
Rhea-Silvia Remus

Canary Islands Winter School, 24.11.2021



# Summary: Computational Methods

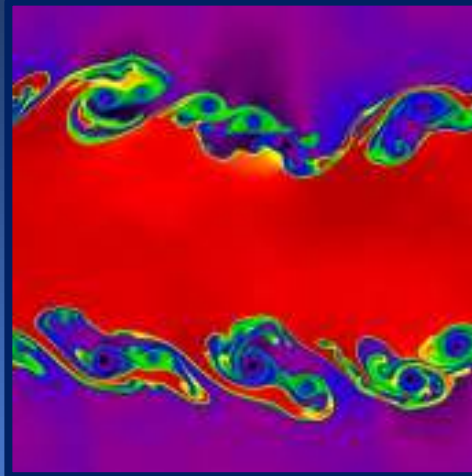
## Smooth Particle Hydrodyn.



Courtesy  
M. Niemeyer, K. Dolag

- ✓ Very good conservation properties (mass, momentum, total energy, angular momentum, entropy)
- ✓ shape invariant
- Instabilities do not grow sufficiently
- Mixing behind shocks not sufficient
- Shocks captured by artificial viscosity

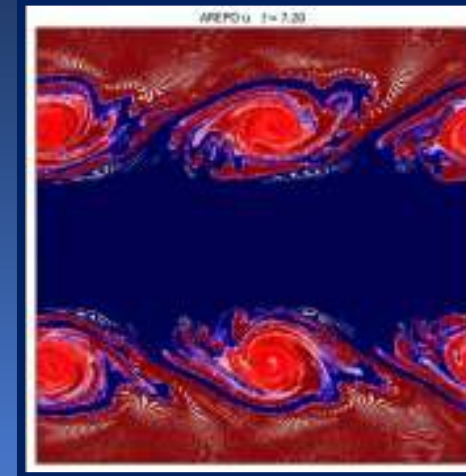
## Adaptive Mesh Refinement



<https://www.astro.princeton.edu/~jstone/Athena/tests/kh/kh.html>

- ✓ Instabilities nicely grow
- ✓ Mixing between phases works well
- Energy conservation issues (especially for fast moving elements)
- Flow over cell boundaries becomes an issue for adaptive meshes
- Not shape invariant

## Moving Mesh



Courtesy  
M. Niemeyer, K. Dolag

- ✓ All good things from the other two
- Flow over cell boundaries (only pseudo-Lagrangian)



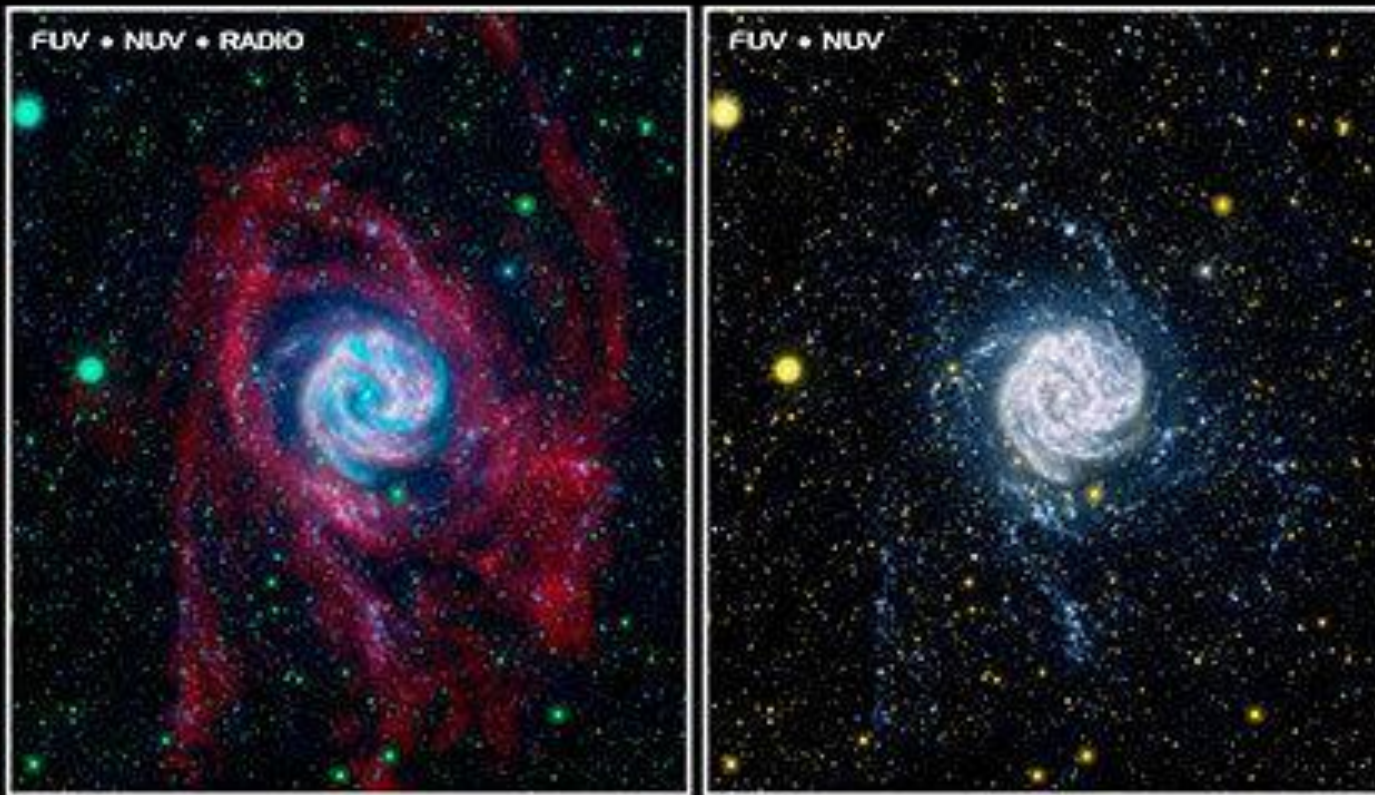
Getting the Universe alight

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# Introduction: It all starts with the gas

## GALEX Galaxy Evolution Explorer



Extended Disk of Galaxy M83

GALEX • NUV • FUV  
VERY LARGE ARRAY • RADIO

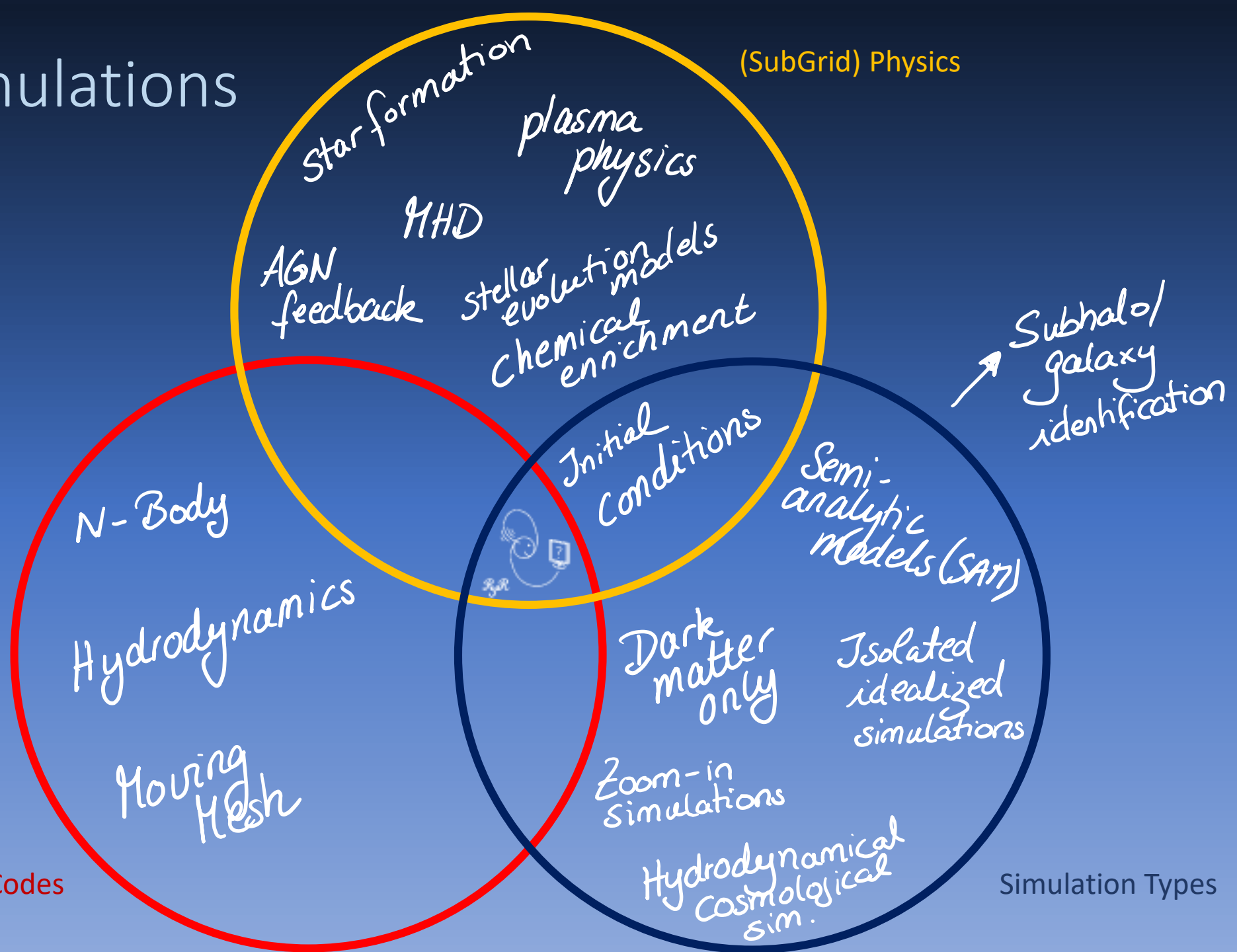
Apart from gravity and the general treatment of gas as either particles (SPH) or fluids (AMR), the physics that affect the baryons, like star formation processes, gas cooling, metal formation, AGN feedback, need to be modelled

# Numerical Simulations

Backbone Codes

(SubGrid) Physics

Simulation Types





# Cooling

For stars to form, gas has to **condense** and **cool** down. This cooling is a function of density and temperature:

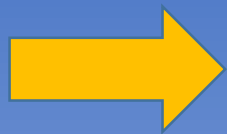
$$\Lambda(T, \rho)$$



# Cooling

## Basic Assumptions

- Optically thin
- Ionization **equilibrium** ( $H, H^+, He, He^+, He^{++}, e^-$ )
- 2-body processes ( $\sim n^2$ )



$$\Lambda(T)/n^2$$





# Cooling

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Cooling happens through:

- Recombination cooling
- Metal line cooling
- Bremsstrahlung cooling/free-free emission
- Compton cooling/heating

$$\Lambda(T)/n^2$$

Lets be more specific:

$$\Lambda(T)/n_e n_H$$

But since  $n_e \approx n_H$

$$\Lambda(T)/n_H^2$$





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- Metal line cooling
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But in fact there are counter processes: **heating** also happens, mostly photoionization heating, but luckily the **cooling usually wins**

$$\Lambda(T)/n^2$$

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$$\Lambda(T)/n_e n_H$$

But since  $n_e \approx n_H$

$$\Lambda(T)/n_H^2$$



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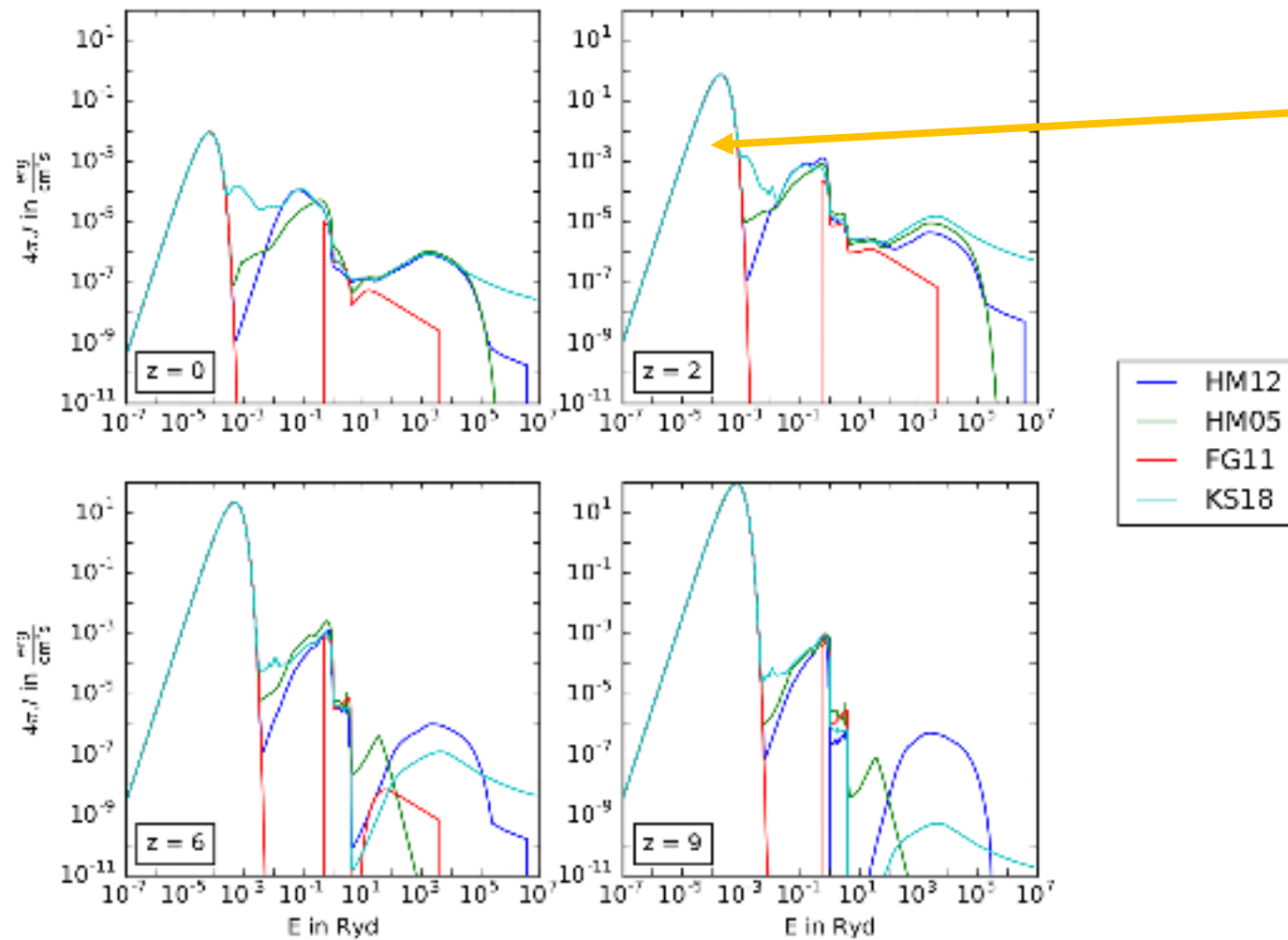
$$\Lambda(T)/n^2$$

This cannot be calculated on the fly! Too many calculation needed!

Thus, **cooling tables** are used



# UV Background



CMB Photons

Credit: S. Lueders



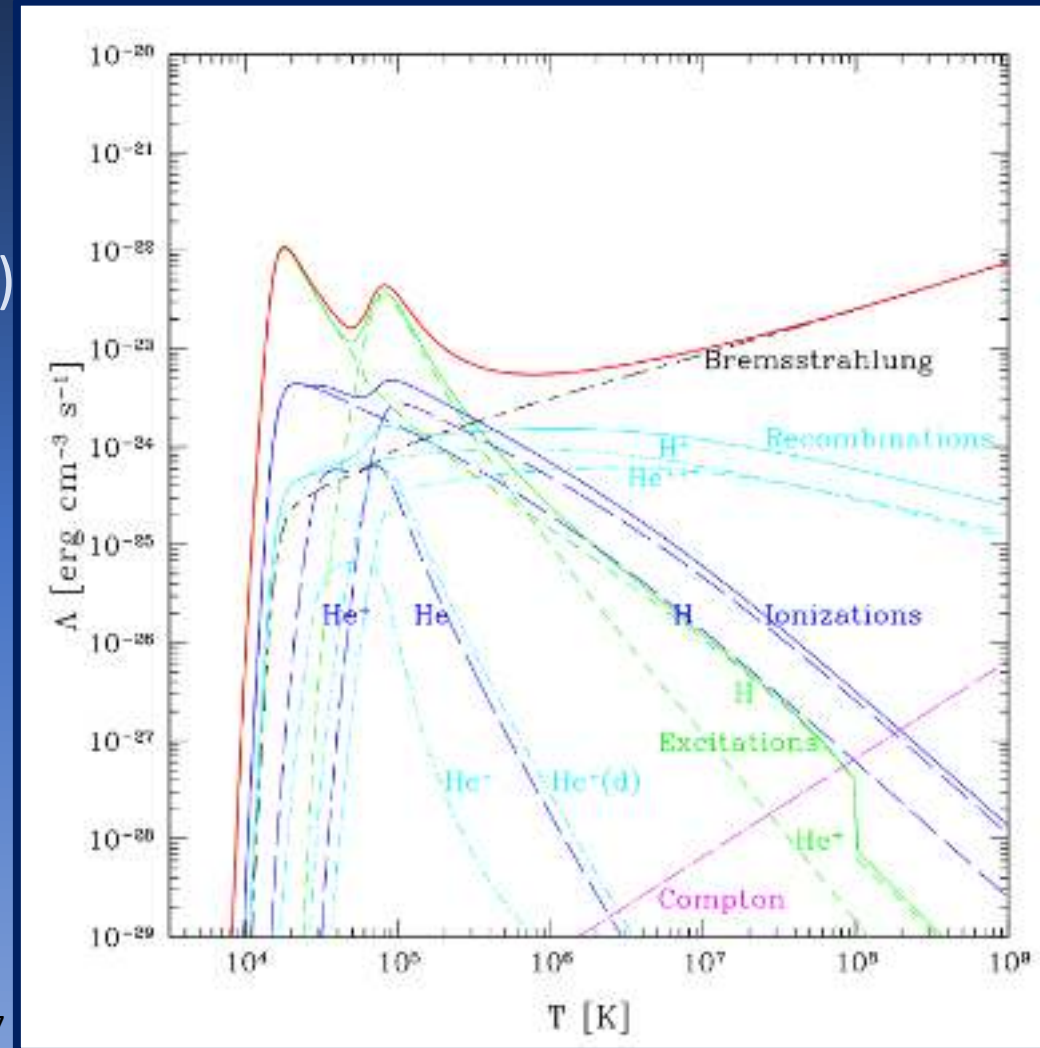
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$$\Lambda(T)/n^2$$

Maio et al., 2007



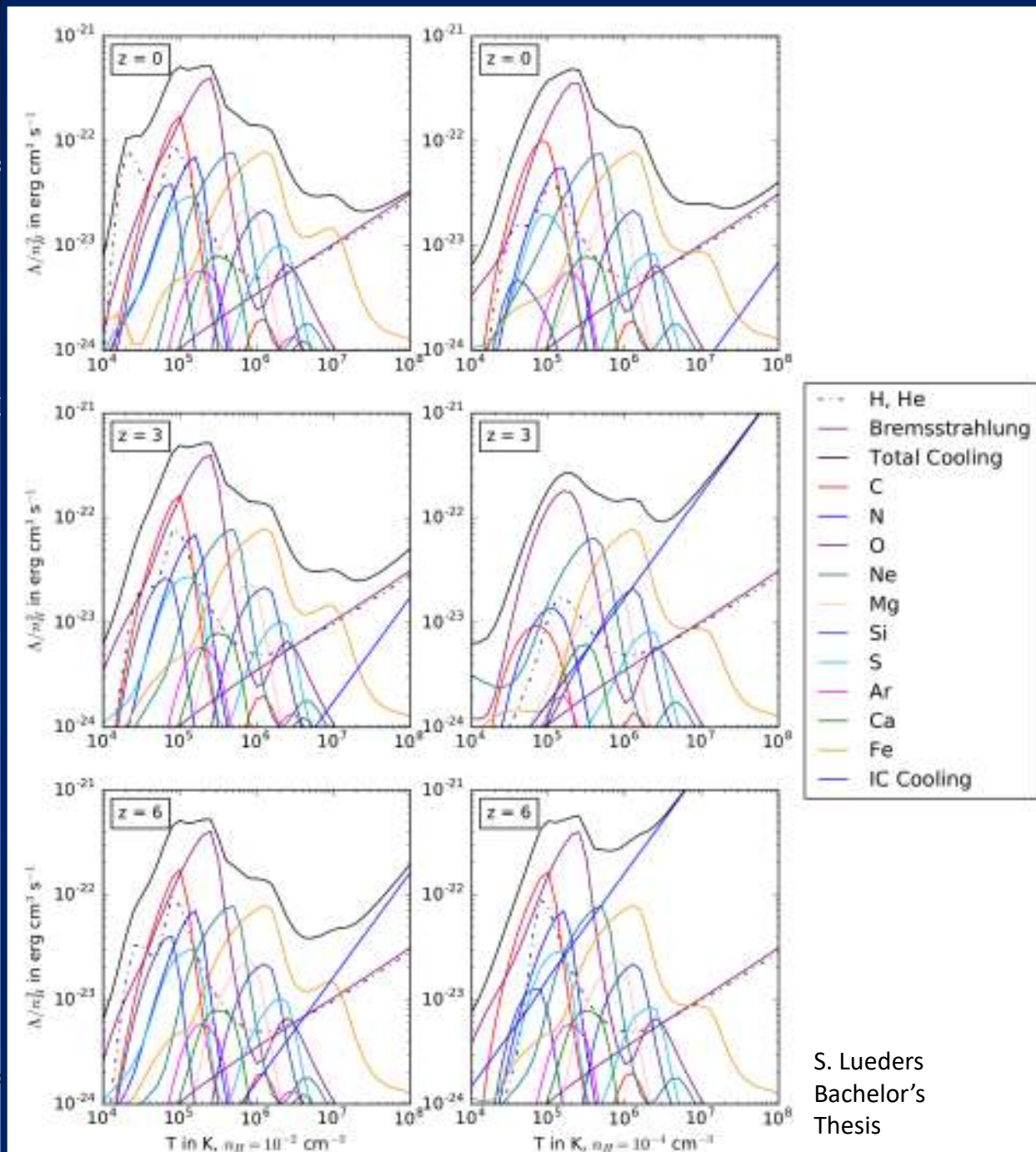


# Cooling

## Basic Assumptions

- Optically thin
- Ionization equilibrium ( $H, H^+, He, He^+, I$ )
- 2-body processes ( $\sim n^2$ )
- Add **Metals** to the Cooling Process

$$\Lambda(T)/n^2$$



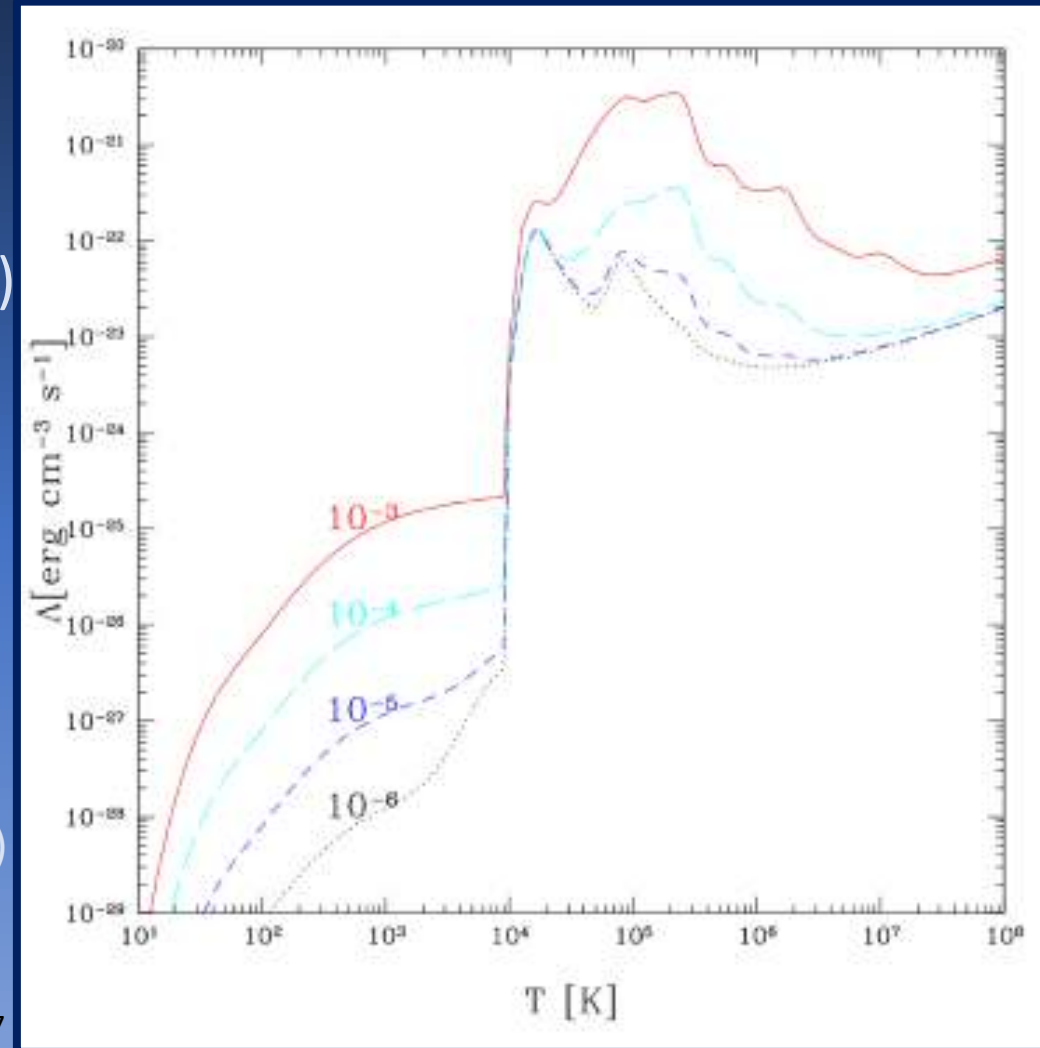


# Cooling

## Basic Assumptions

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- 2-body processes ( $\sim n^2$ )
- Add Metals to the Cooling Process
- Below  $T \sim 10^4 K$ :
  - Solving balance equations
  - Cooling by molecules that need to be traced ( $H_2, HD, \dots$ )
  - Plus fine-structure transitions in metals (FeII, OI, SiII, CII...)

Maio et al., 2007





# Cooling

## Basic Assumptions

- Optically thin

- Ionization state:  $\text{H} \rightleftharpoons \text{H}^+ \quad \text{He} \rightleftharpoons \text{He}^+ \quad \text{He}^{++} \rightleftharpoons \text{He}^{+++} \rightleftharpoons \dots$

- 2

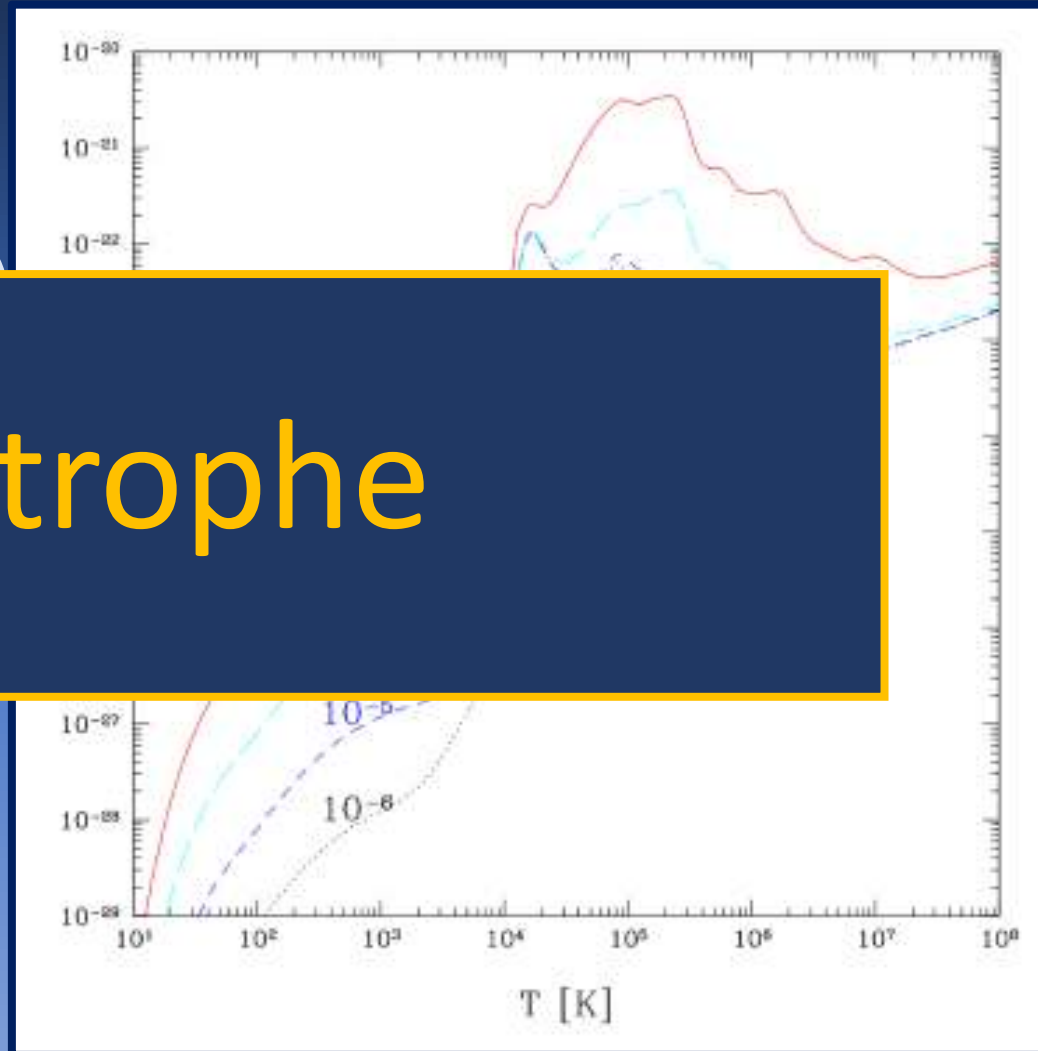
- A

- E

- Solving balance equations
- Cooling by molecules that need to be traced ( $\text{H}_2$ ,  $\text{HD}$ , ...)
- Plus fine-structure transitions in metals ( $\text{FeII}$ ,  $\text{OI}$ ,  $\text{SiII}$ ,  $\text{CII}$ ...)

$$\Lambda(T)/n^2$$

Maio et al., 2007







# Star Formation

In nature, cooled gas will form stars → on scales way below our resolution!



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## Basic Assumptions

- Convergent flow:  $\nabla v < 0$
- High density region:  $\rho > 0.1 \text{ Atoms/cm}^3$
- Jeans instability:  $\lambda/c > t_{dyn}$  with  $t_{dyn} = (4\pi G\rho)^{-0.5}$



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➤ **Conversion** of cold gas into stars:  $\frac{d\rho_*}{dt} = -\frac{d\rho}{dt} = \frac{c_*\rho}{t_*};$

$c_*$ : star formation efficiency ( $\approx 10\%$ );  $t_*$ : star formation time (max. of  $t_{dyn}$  and  $t_{cool}$ )



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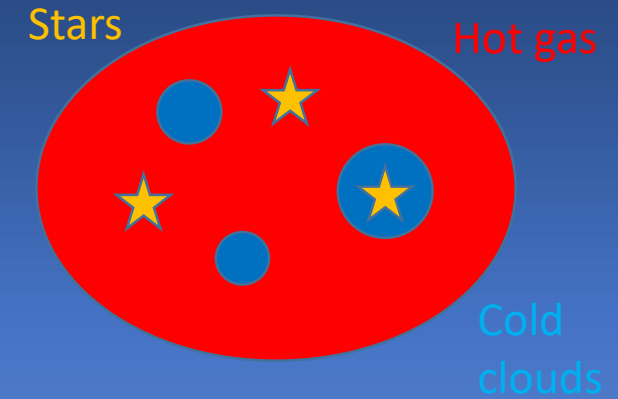
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- Heating of the gas by Type-II Supernovae ( $10^{51}\text{erg/SN}$ ). Number of SNII from IMF
- When a significant fraction of gas is converted into stars, **spawn new star** particles



# Star Formation

## More elaborated: a multi-phase model

Springel & Hernquist 2002



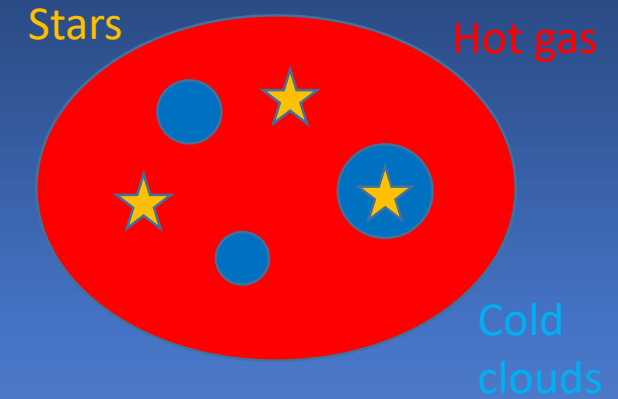


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Springel & Hernquist 2002

- Cold clouds  $\rho_c$  in **pressure equilibrium** with hot gas  $\rho_h$







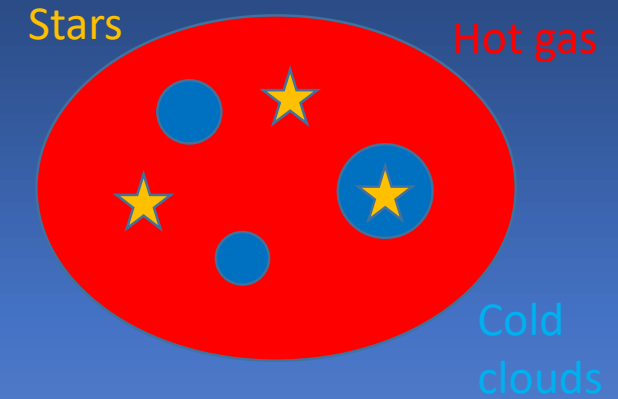
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Cooling function





# Star Formation

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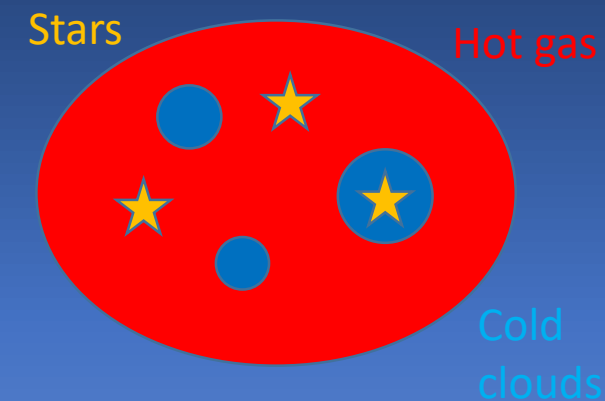
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- Cold clouds **form stars**:  $\frac{d\rho_*}{dt} = \frac{\rho_c}{t} - \beta \frac{\rho_c}{t_*}$

Star formation timescale

Supernova mass  
fraction





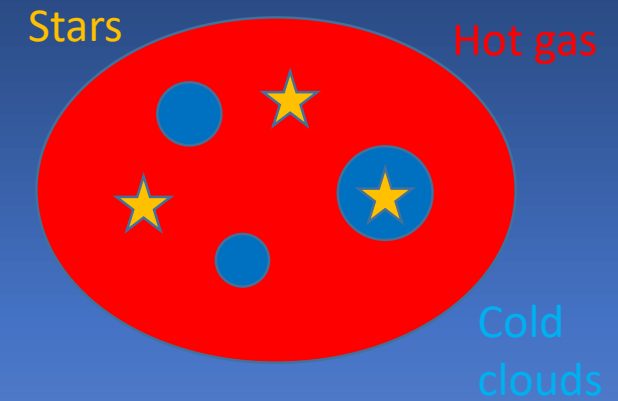
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- Cold clouds form stars:  $\frac{d\rho_*}{dt} = \frac{\rho_c}{t} - \beta \frac{\rho_c}{t_*}$
- Cloud **evaporation**:  $\frac{d\rho_h}{dt} = A\beta \frac{\rho_c}{t_*}$  through heating from SN-II

Cloud evaporation  
parameter



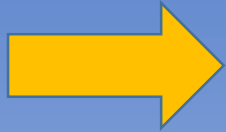
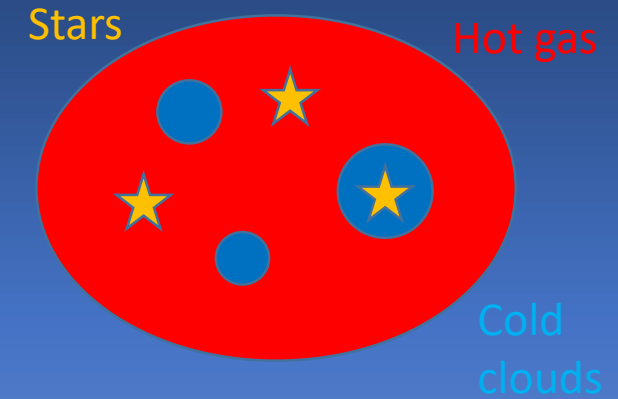


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Self-regulated star-formation, but a complex set of differential equations needs to be solved



# Star Formation

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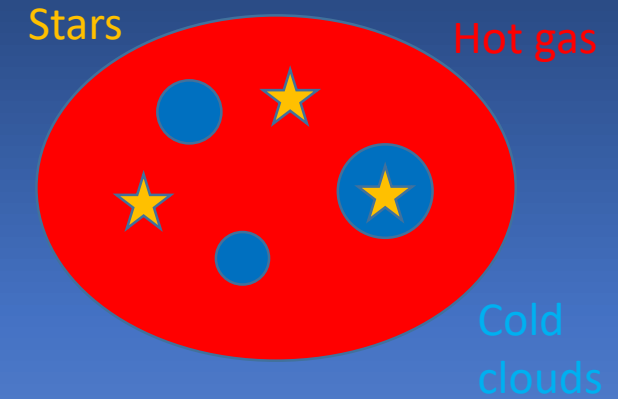
Springel & Hernquist 2002

Self-regulated star-formation, but a complex set of differential equations needs to be solved

At the end, there are three free parameters left to adjust: the thermal instability density parameter, an evaporation efficiency parameter, and the star formation timescale. First two can be constrained by assumptions regarding the temperature range of the ISM.

Star formation timescale can only be adjusted using

**Kennicutt-law** (Kennicutt 1998)



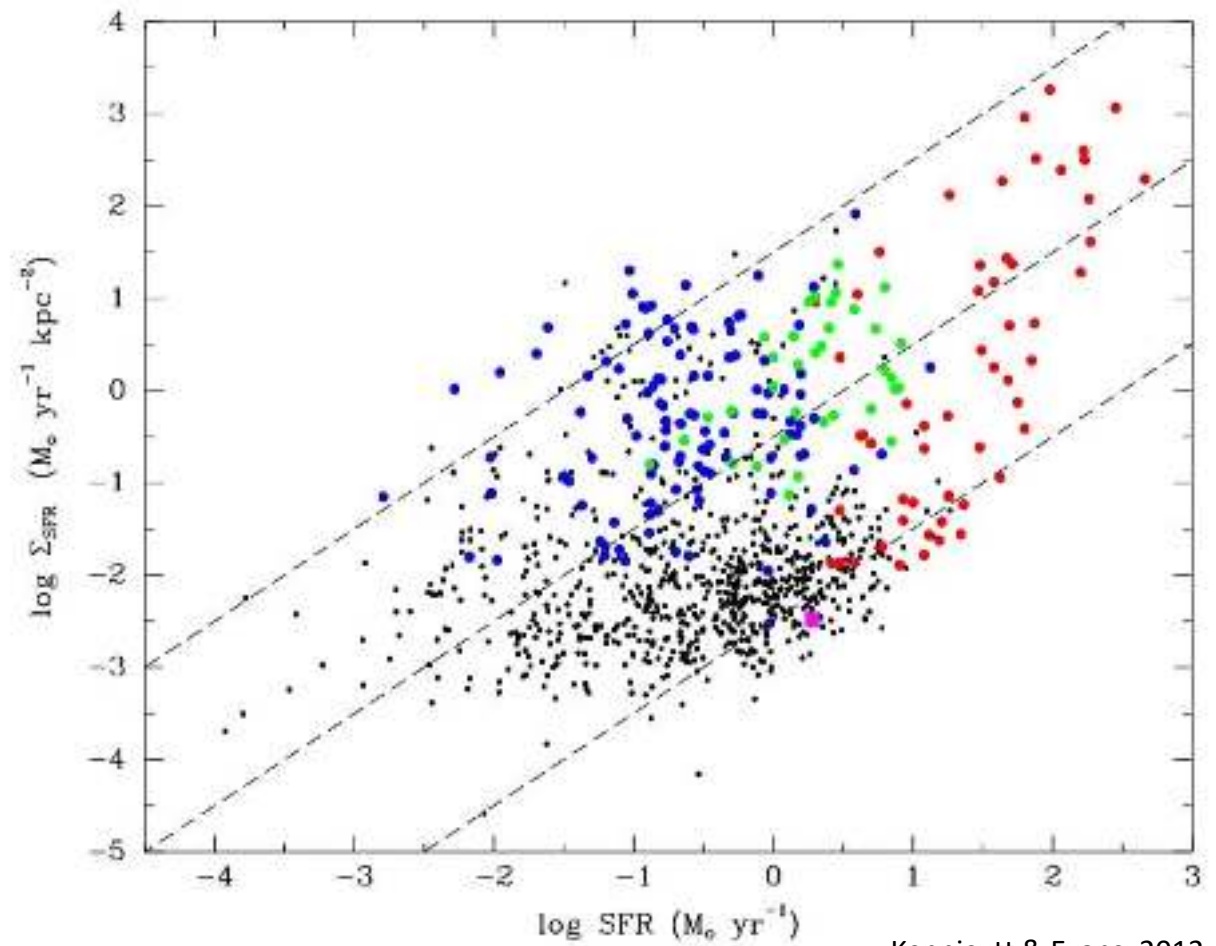


# Star Formation

Star formation timescale can only be adjusted using Kennicutt-law

(Kennicutt 1998)

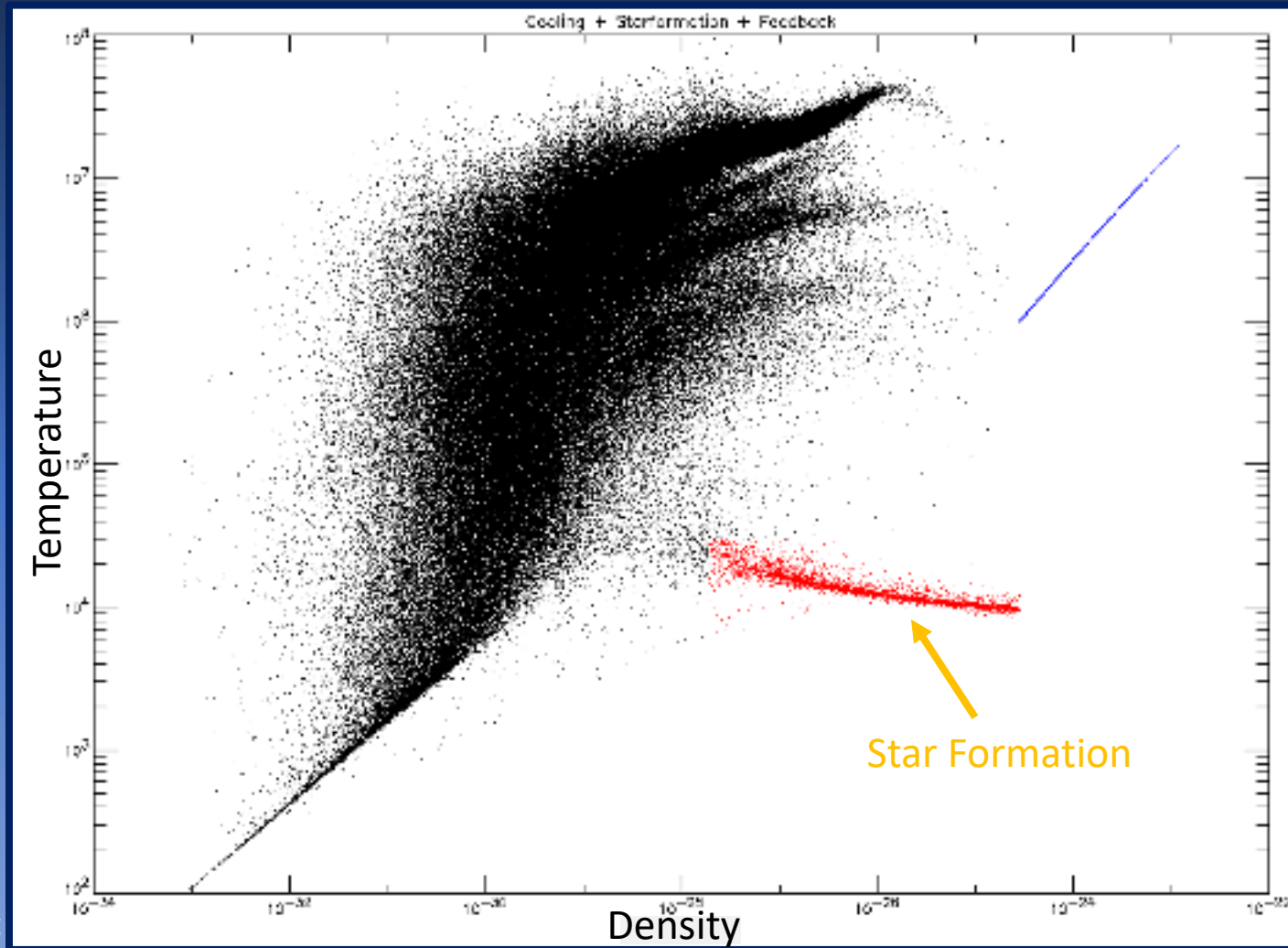
$$\Sigma_{\text{SFR}} = (2.5 \pm 0.7) \times 10^{-4} \left( \frac{\Sigma_{\text{gas}}}{\text{M}_{\odot} \text{pc}^{-2}} \right)^{1.4 \pm 0.15} \frac{\text{M}_{\odot}}{\text{yr kpc}^2}$$



Kennicutt & Evans, 2012



# Star Formation



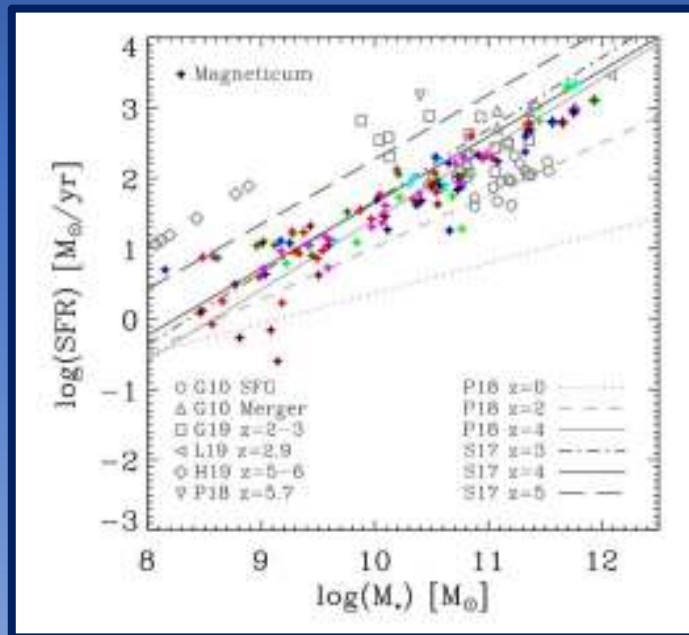
Credit: Klaus Dolag



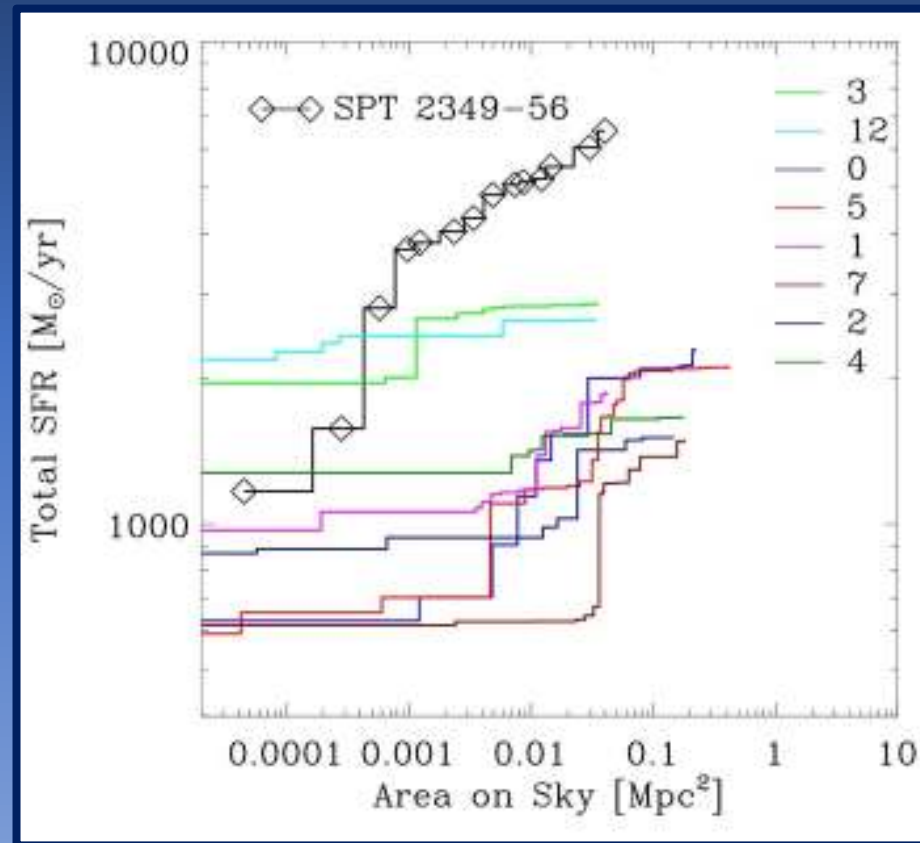
# Star Formation

But:

Fail to reproduce high star formation rates seen at high redshifts, for example in proto clusters



Remus et al., 2022



SPT 2349-56:  
Miller et al., 2018





# Star Formation

## How to improve?

- Schmitt-Kennicutt assumption is not correct for higher redshifts
- Star formation efficiency should be coupled to molecular gas, not cold gas in general



# Star Formation

## How to improve?

- Schmitt-Kennicutt assumption is not correct for higher redshifts
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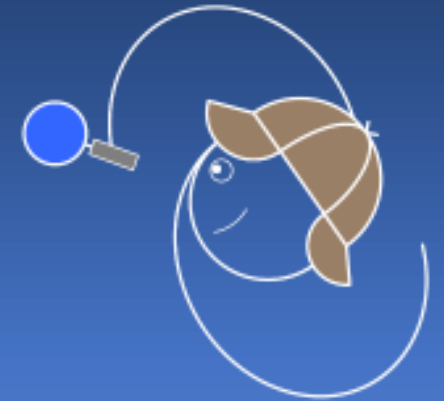


$$\frac{d\rho_*}{dt} = (1 - \beta) \frac{\rho_c}{t_*}$$

Replace  $\rho_c$  by a mass loading that is coupled to the molecular hydrogen mass:

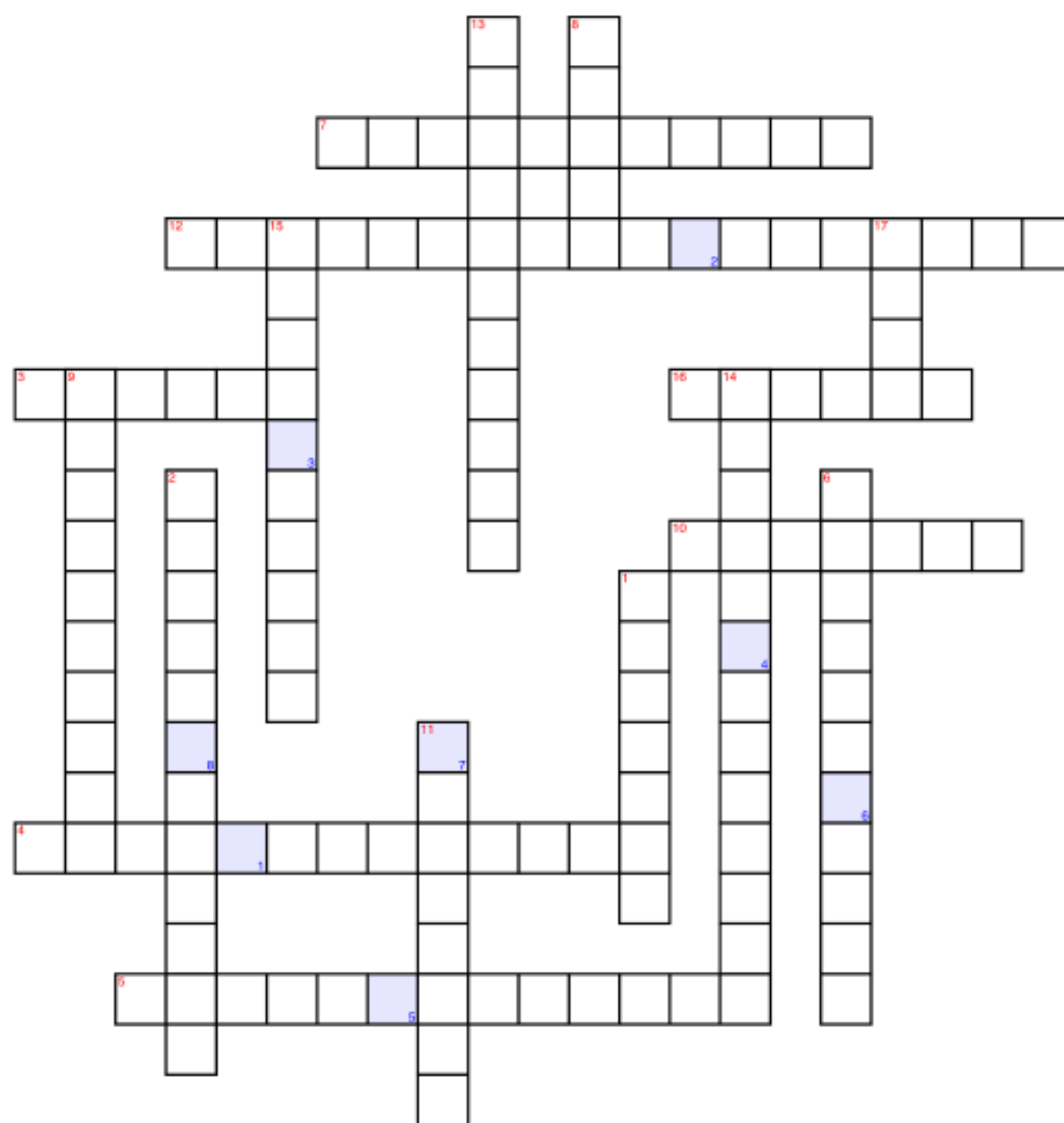
$$\dot{M}_{SF} = f_* \frac{M_{H_2}}{t_{dyn}}$$

# Take a Break: Riddle



1. Necessary for gas to be able to form stars
2. Drives ionization in the early universe
3. Everything not hydrogen or helium
4. Basic process that transforms gas
5. Thermal, Kelvin–Helmholtz, Rayleigh–Taylor
6. Radiation from (hot) stars causes cloud \_\_\_\_\_
7. What star particles and countries have in common
8. A criterion for collapse but also a fashion item
9. Happens to gas through stellar death and at the stock market
10. The force that dropped an apple onto Newton
11. An IMF
12. Diffuse component in a galaxy cluster
13. Time-independent predictions are only possible for \_\_\_\_\_
14. Basic physics that describes baryons
15. To partition a plane into regions with certain properties
16. Caused by high-Mach-number flows
17. Most of the matter is like this

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|





# Feedback

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# Feedback

Feedback comes from **two** different sources:

- 1) Massive Stars and Supernovae
- 2) Supermassive Black Holes (AGN)



# Feedback: Supernovae

Our stellar particles are not just single **stellar particles** but rather conglomerates of stars. Every time a star is born, it actually is a **whole population of stars**.



# Feedback: Supernovae

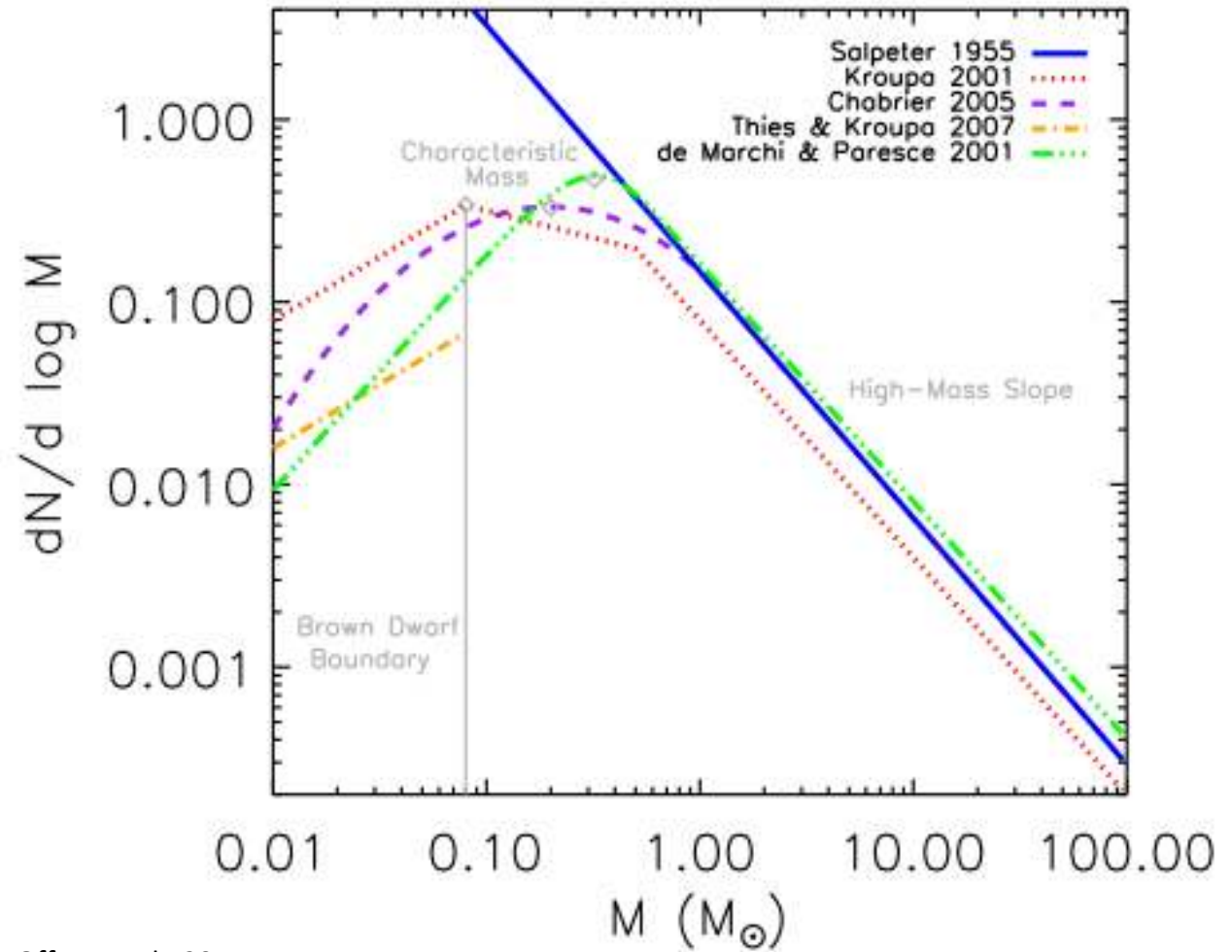
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Here, we use an **initial mass function** (IMF) to emulate the population



# Feedback: Supernovae

Our stellar particles are not just single stellar particles but rather conglomerates of stars. Every time a star is born, it actually is a whole population of stars. Here, we use an IMF to emulate the population

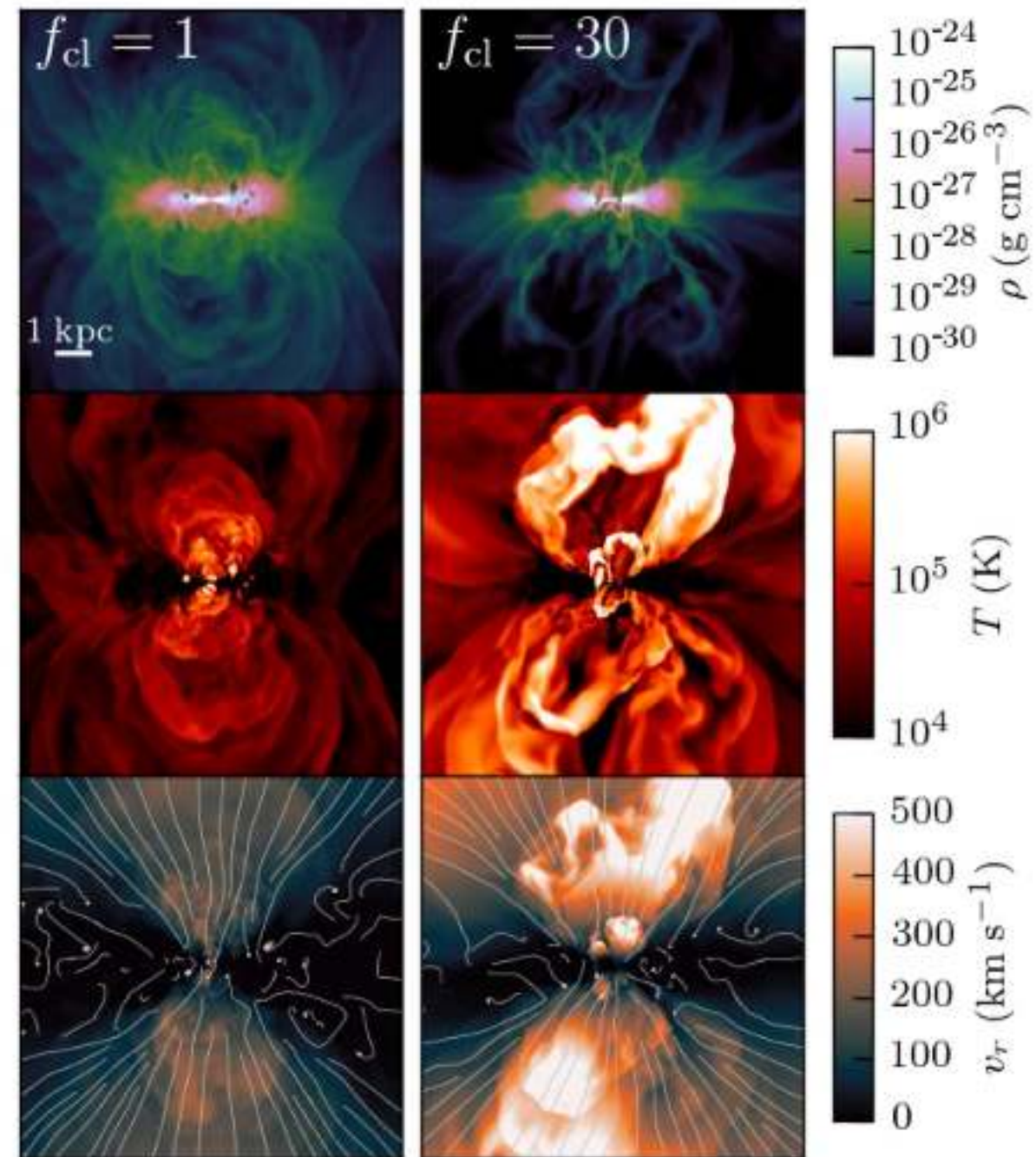


Offner et al., 2014





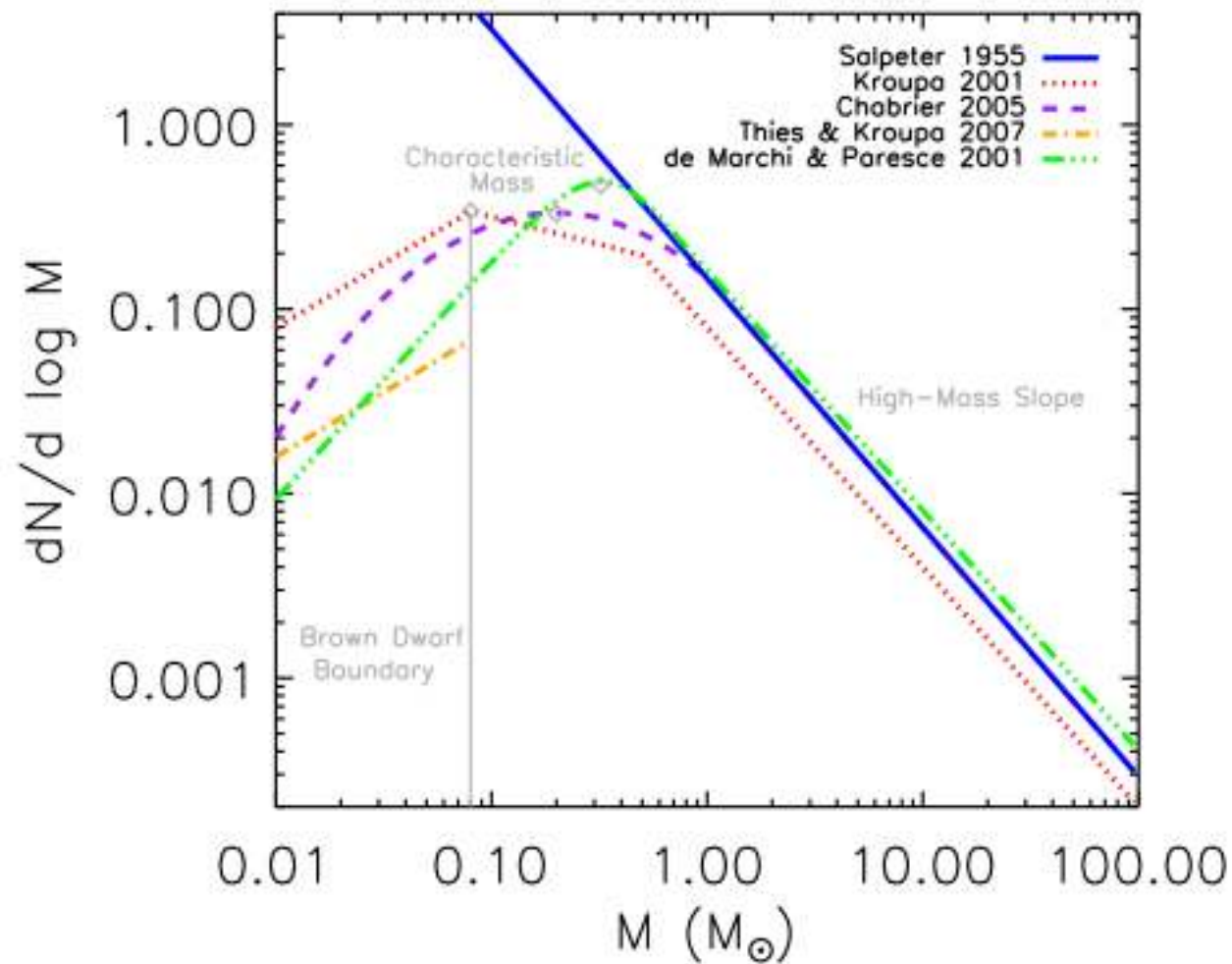
# Feedback: Supernovae





# Feedback: Supernovae

But not just Supernovae, also **massive stars** have winds that drive (kinetic) feedback



Offner et al., 2014



# Feedback: AGN

The existence of a Black Hole is often revealed by a jet that can be observed, as for example in M87, the second brightest galaxy of the Virgo cluster.

M87



Image credit: NASA, ESA and the Hubble Heritage Team (STScI/AURA);  
Acknowledgment: P. Cote (Herzberg Institute of Astrophysics) and E. Baltz (Stanford University)



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**Credit:** EHT Collaboration

M87

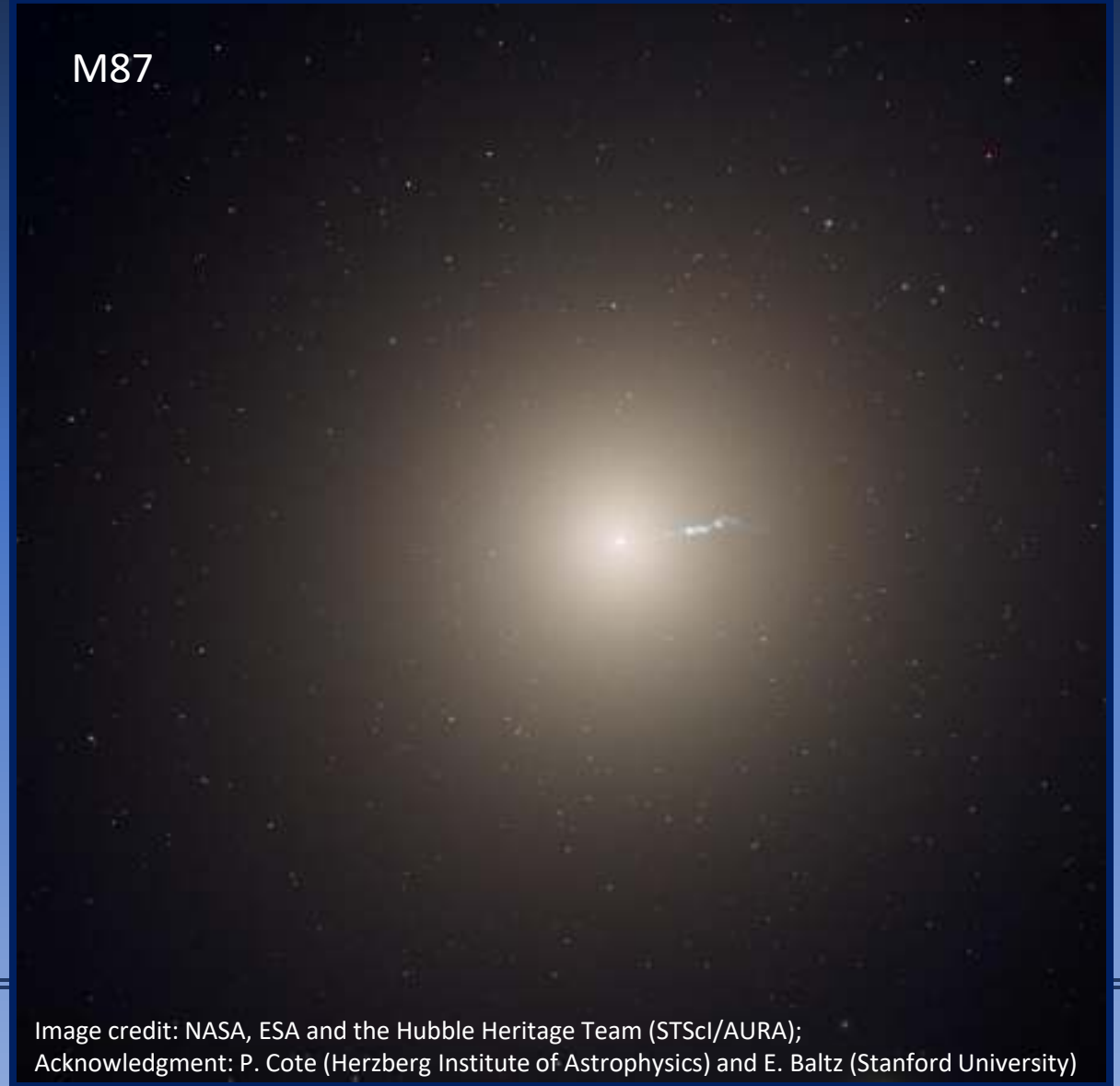


Image credit: NASA, ESA and the Hubble Heritage Team (STScI/AURA);  
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# Feedback: AGN

Black Holes are included in the simulations as **sink particles**



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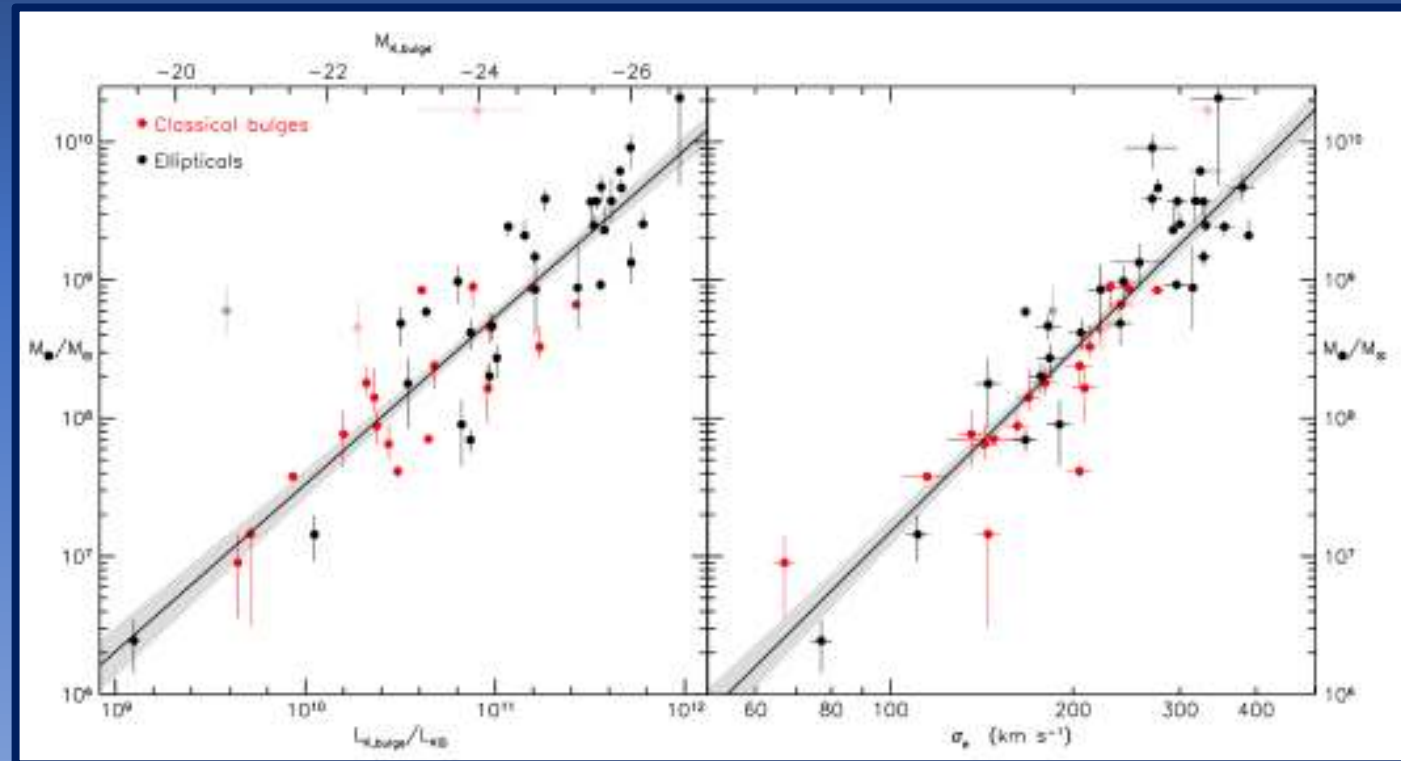


# Feedback: AGN

Black Holes are included in the simulations as sink particles



Either seeding at a constant mass, or on the  $m_{BH}$ - $\sigma$  relation

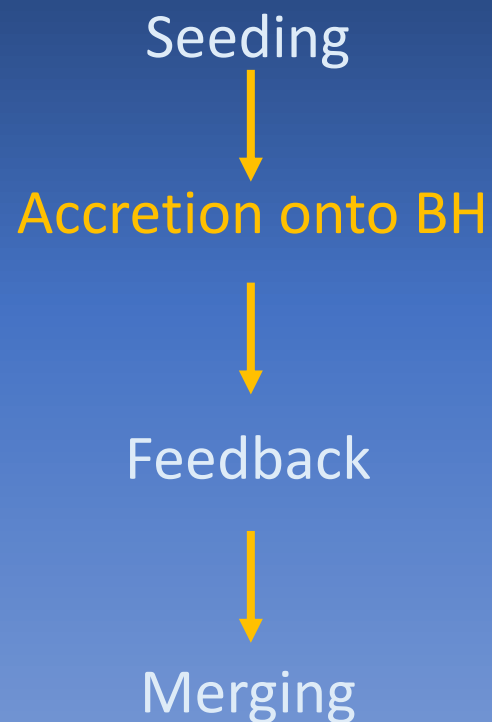


Kormendy & Ho 2013



# Feedback: AGN

Black Holes are included in the simulations as sink particles



Bondi Accretion:

$$\dot{M}_B = \frac{4\pi\alpha G^2 M_{BH}^2 \rho}{(c_s^2 + v^2)^{3/2}},$$

Gas density

Sound speed

Eddington Limit:

$$\dot{M}_{Edd} \equiv \frac{4\pi G M_{BH} m_p}{\epsilon_r \sigma_T c},$$

Implemented:

$$\dot{M}_{BH} = \min(\dot{M}_{Edd}, \dot{M}_B).$$





# Feedback: AGN

Black Holes are included in the simulations as sink particles



$$L_{bol} = 0.1 \dot{M}_{BH} c^2$$

$$\dot{E}_{feedback} = f L_{bol}$$

efficiency



# Feedback: AGN

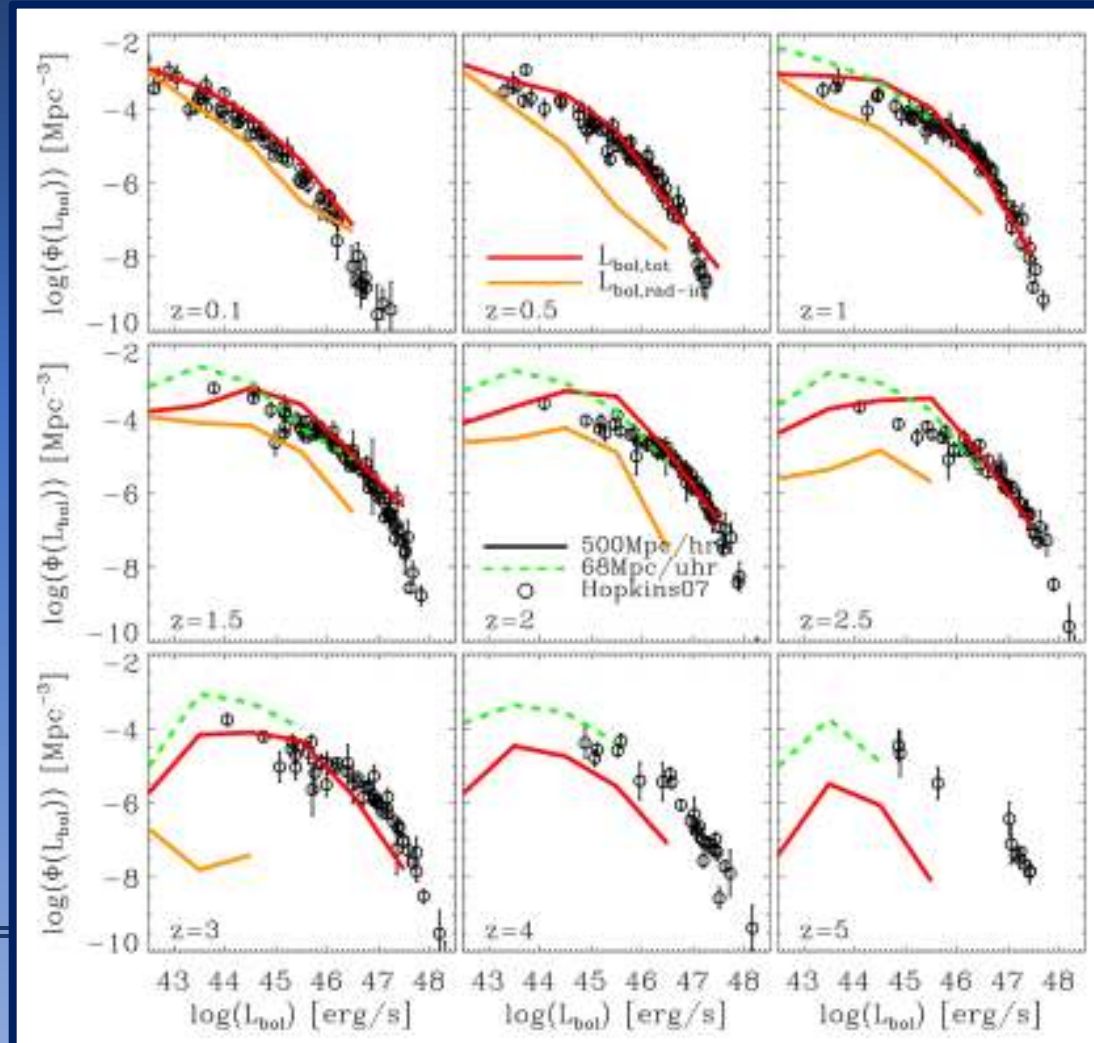
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Hirschmann et al., 2014





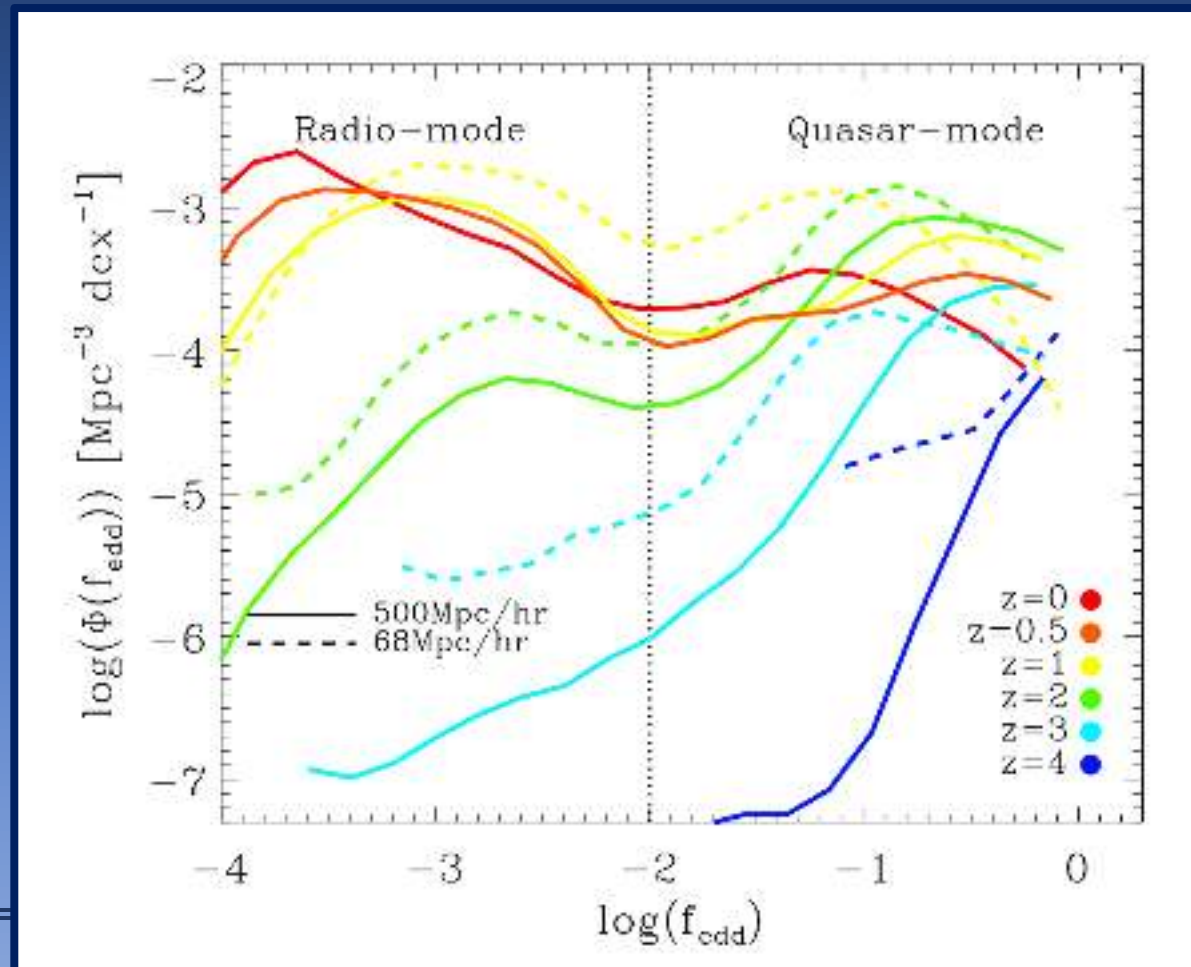
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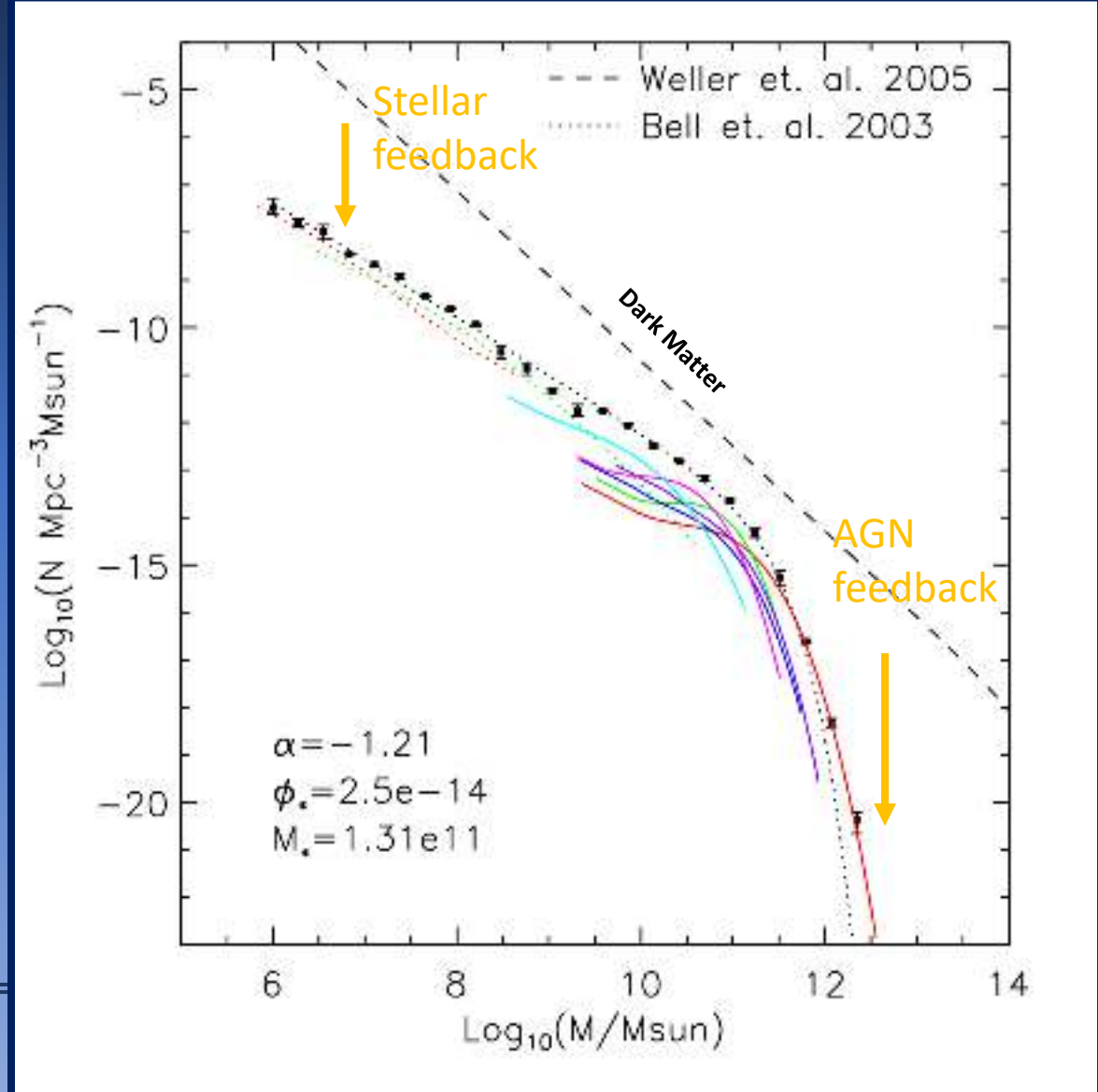


Pinning or no pinning?

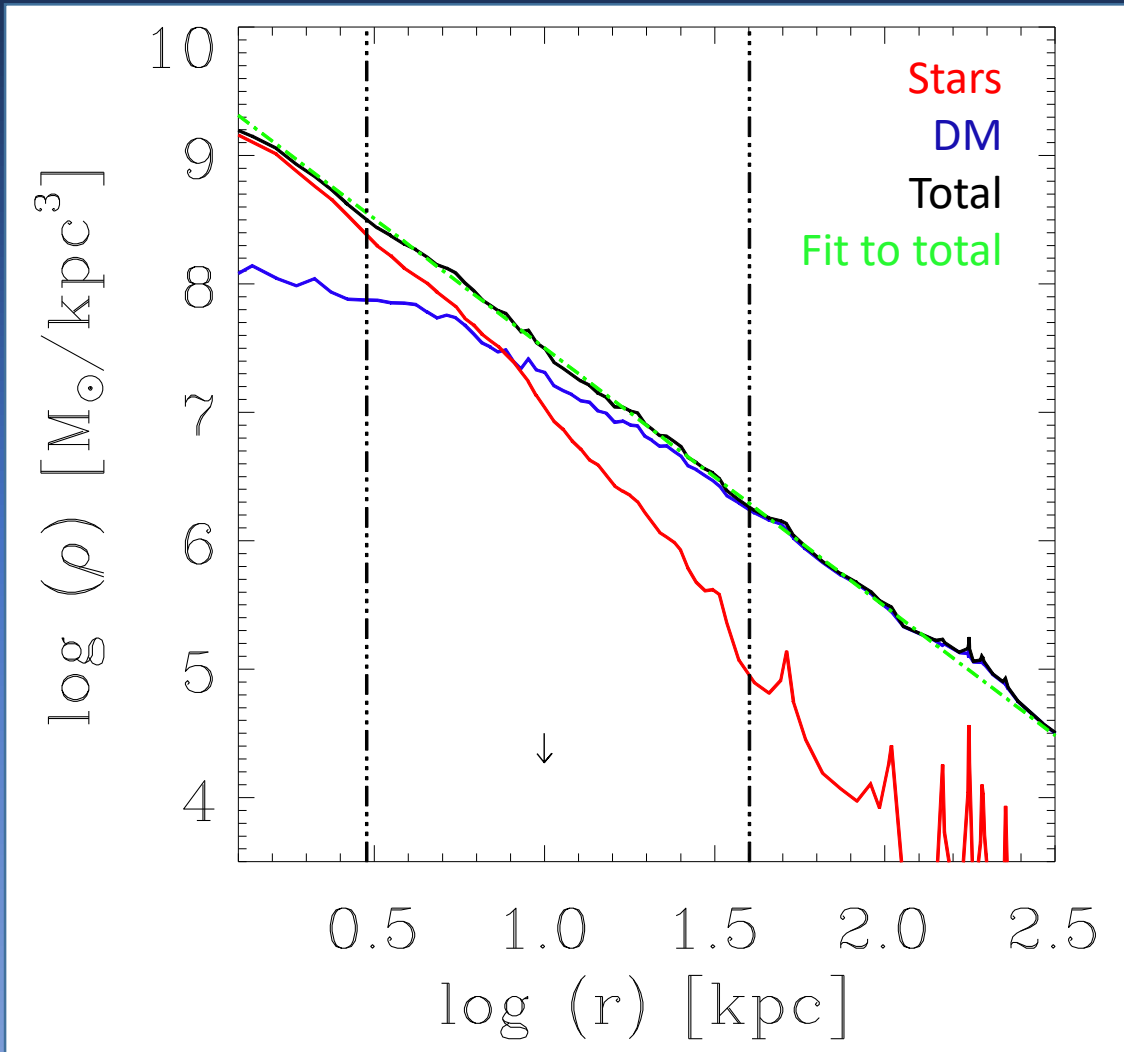


# Feedback

How feedback influences galaxies



# Radial Density Profiles



Total radial density profiles can be fit by a single power law.

Inner part: Stars dominate the total profiles.

Outer part: Dark Matter dominates the total profiles.

$$\gamma = \frac{d \log(\rho)}{d \log(r)}$$

Most ETGs have slopes close to isothermal, i.e.  $\gamma_{\text{tot}} \approx -2$ , but they can be as steep as  $\gamma_{\text{tot}} \approx -3$ .



This is independent of the included feedback models

see Remus et al., 2013; 2017

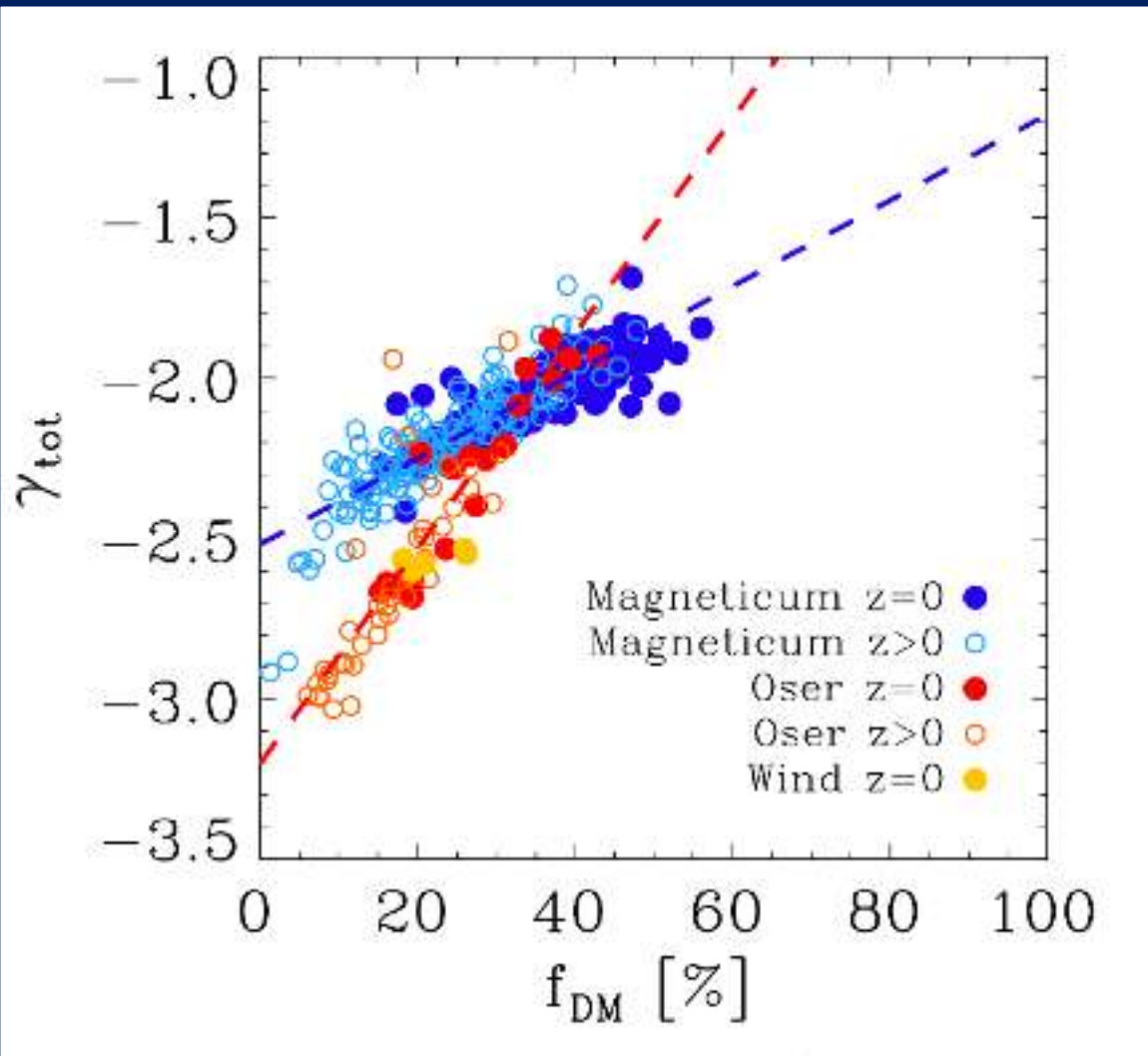
# The Role of Feedback

To understand the impact of the different feedback models on the implementation of these scaling relations, we use ETGs from simulations with different feedback models:



|            | SPH      | AGN | Stellar Wind | Cooling      | References                                  |
|------------|----------|-----|--------------|--------------|---------------------------------------------|
| Magneticum | Improved | Yes | Weak         | Incl. metals | Hirschmann et al., 2014; Teklu et al., 2015 |
| Oser       | Standard | No  | No           | Primordial   | Oser et al., 2010; 2012                     |
| Wind       | Standard | No  | Strong       | Incl. metals | Hirschmann et al., 2013; 2015               |

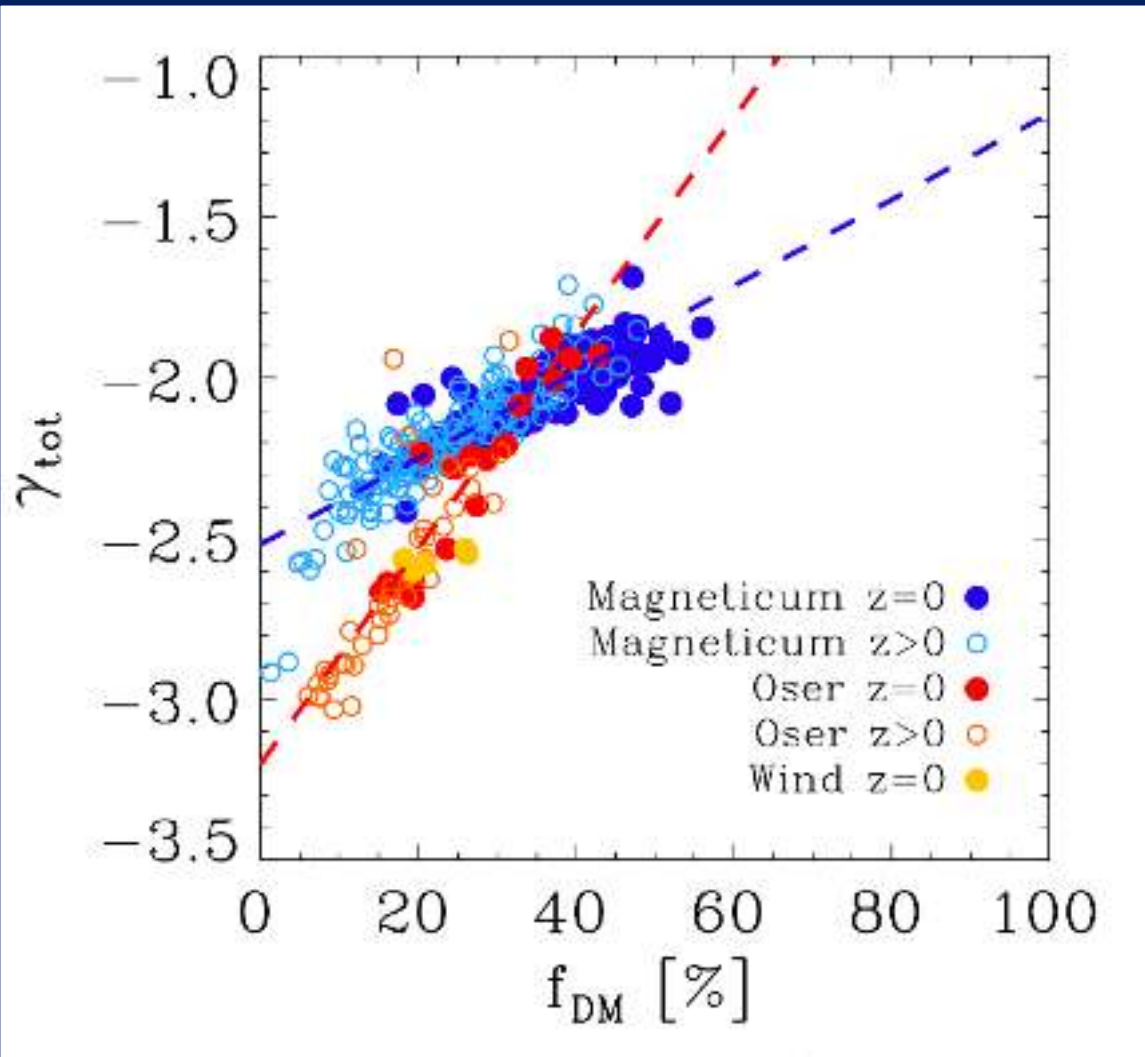
# Central Dark Matter Fractions



The fraction of dark matter within the halfmass radius is lower at higher redshifts, and it is strongly correlated with the slope of the total density profiles.



# Central Dark Matter Fractions

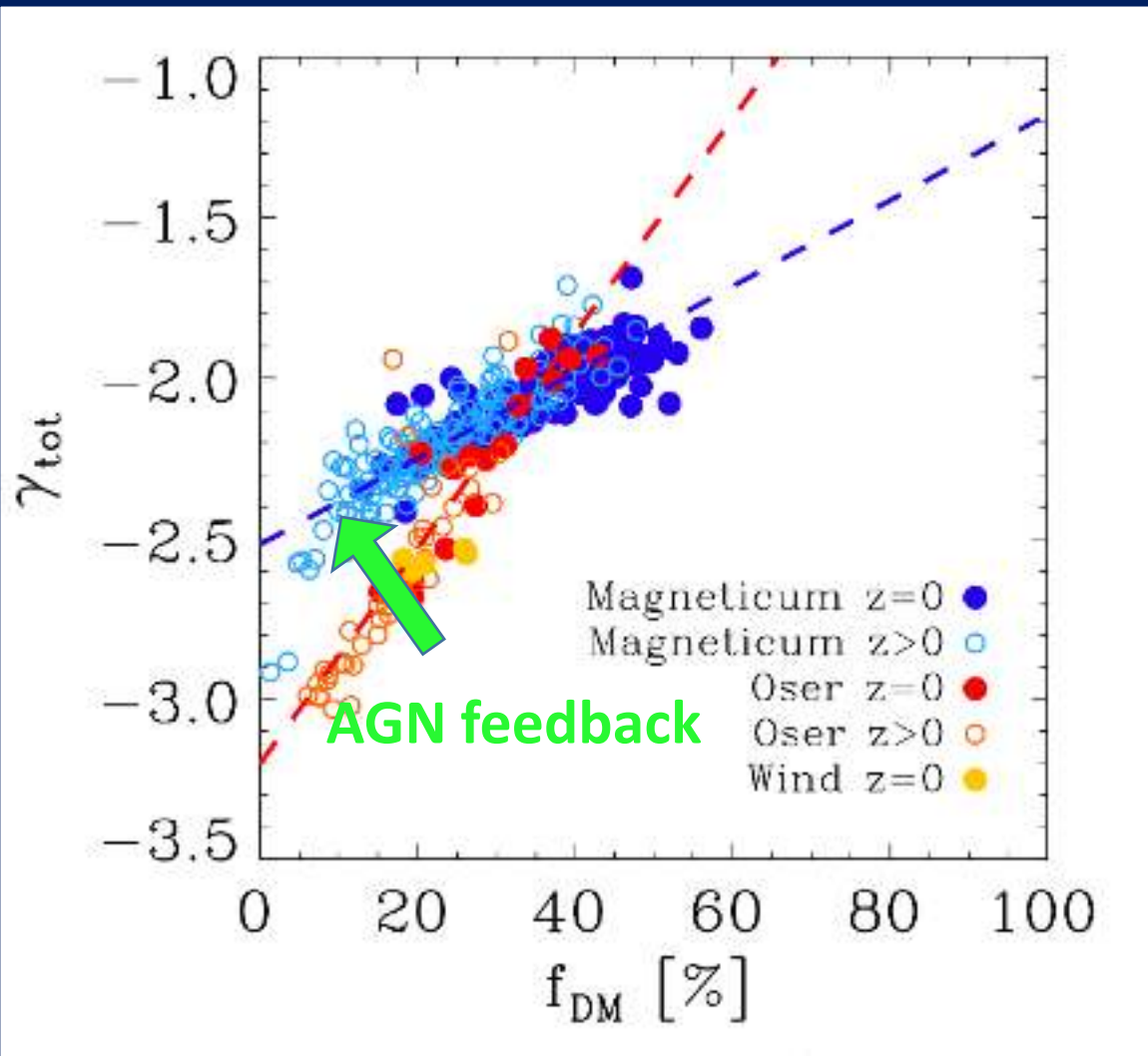


The fraction of dark matter within the halfmass radius is lower at higher redshifts, and it is strongly correlated with the slope of the total density profiles.



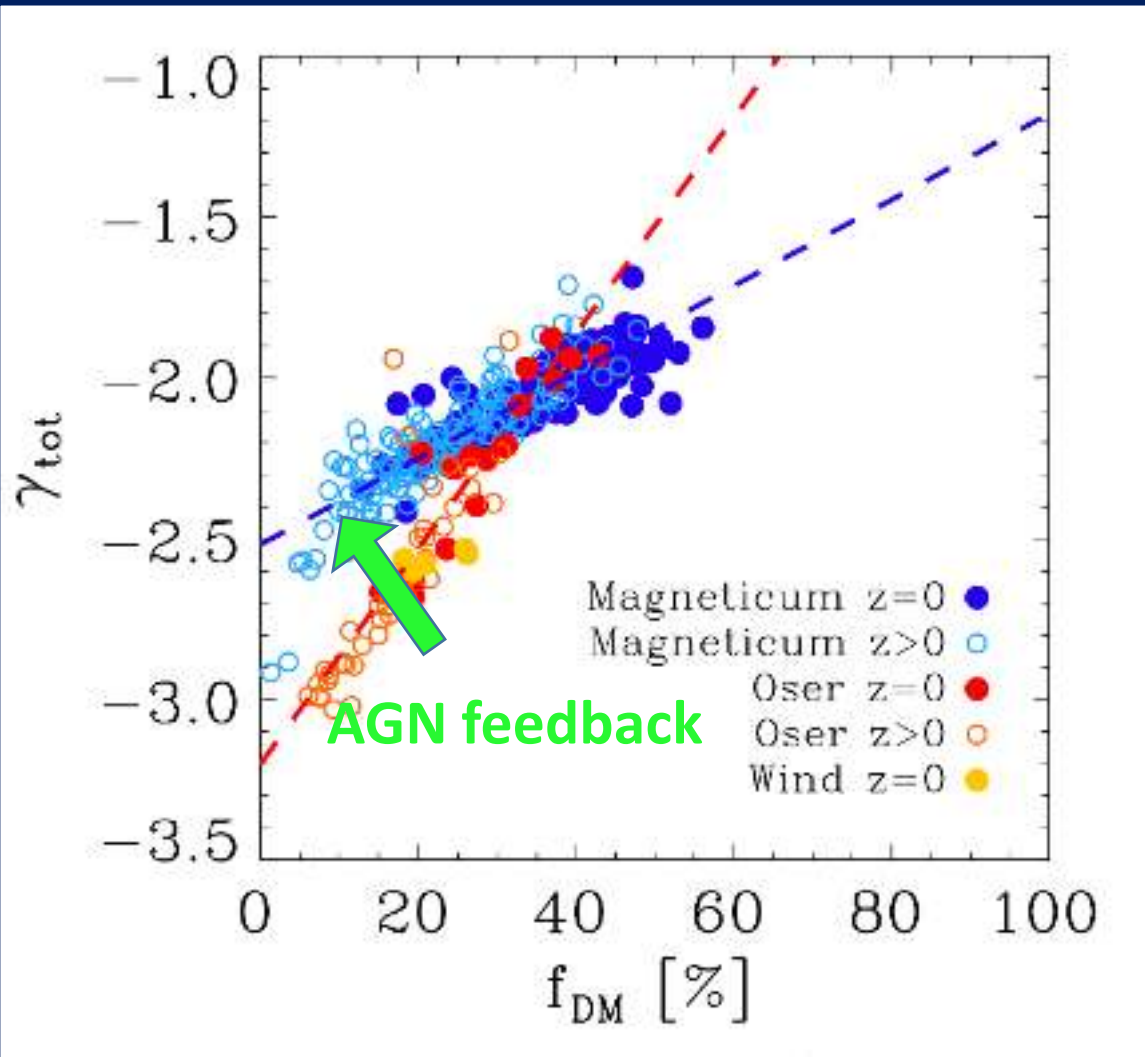
The correlation between the dark matter fraction and the total density slope already establishes at high redshifts

# Central Dark Matter Fractions



The fraction of dark matter within the halfmass radius is lower at higher redshifts, and it is strongly correlated with the slope of the total density profiles.

# Central Dark Matter Fractions

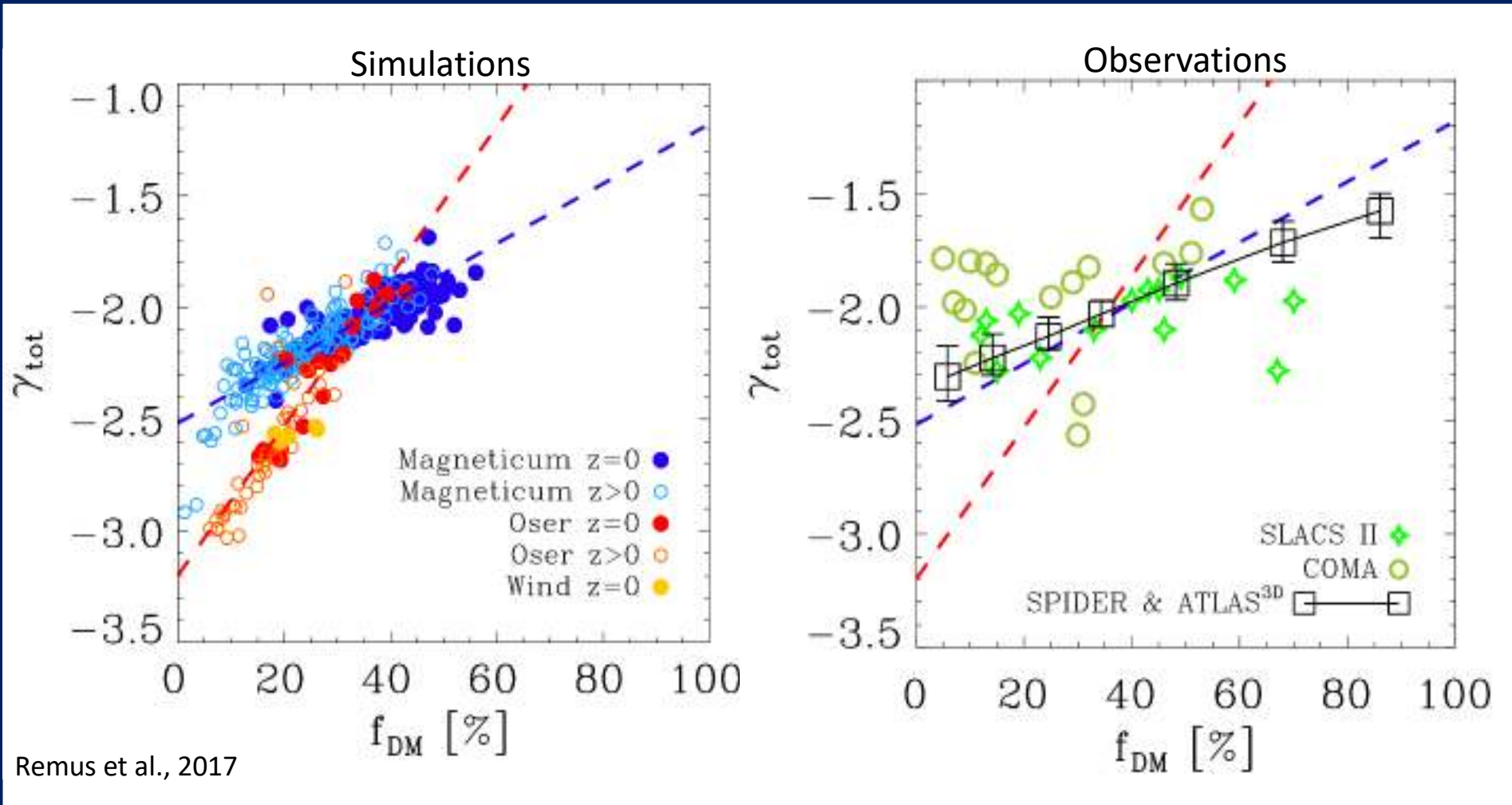


The fraction of dark matter within the halfmass radius is lower at higher redshifts, and it is strongly correlated with the slope of the total density profiles.



The slope of this relation can be used to test the different feedback models

# Co-Evolution of Dark Matter Fractions and Density Profiles



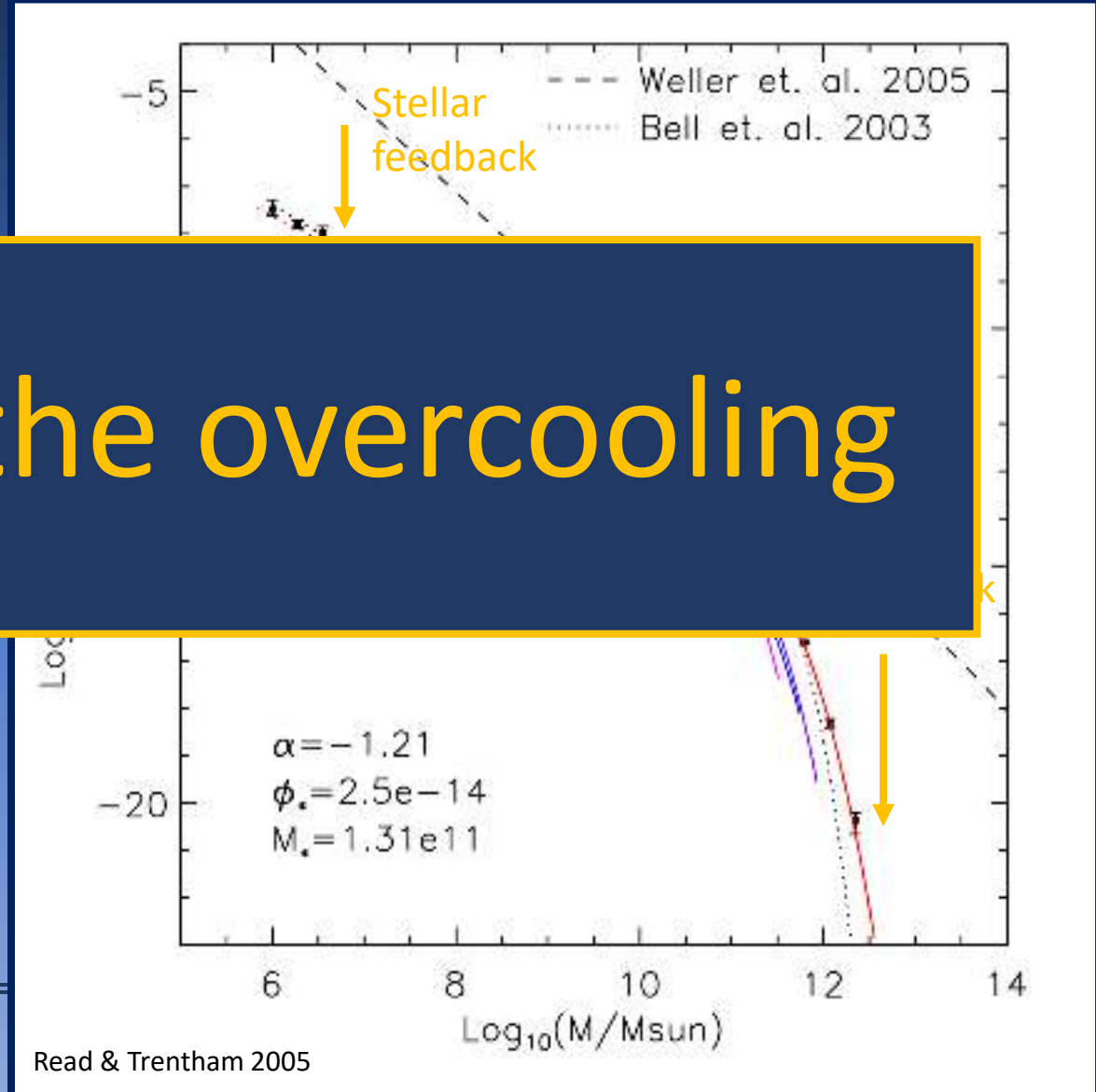
Observations:  
SPIDER & ATLAS<sup>3D</sup> : Tortora et al., 2014  
Coma: Thomas et al., 2007  
SLACS: Barnabé et al., 2011



# Feedback

How feedback influences galaxies

Feedback stops the overcooling

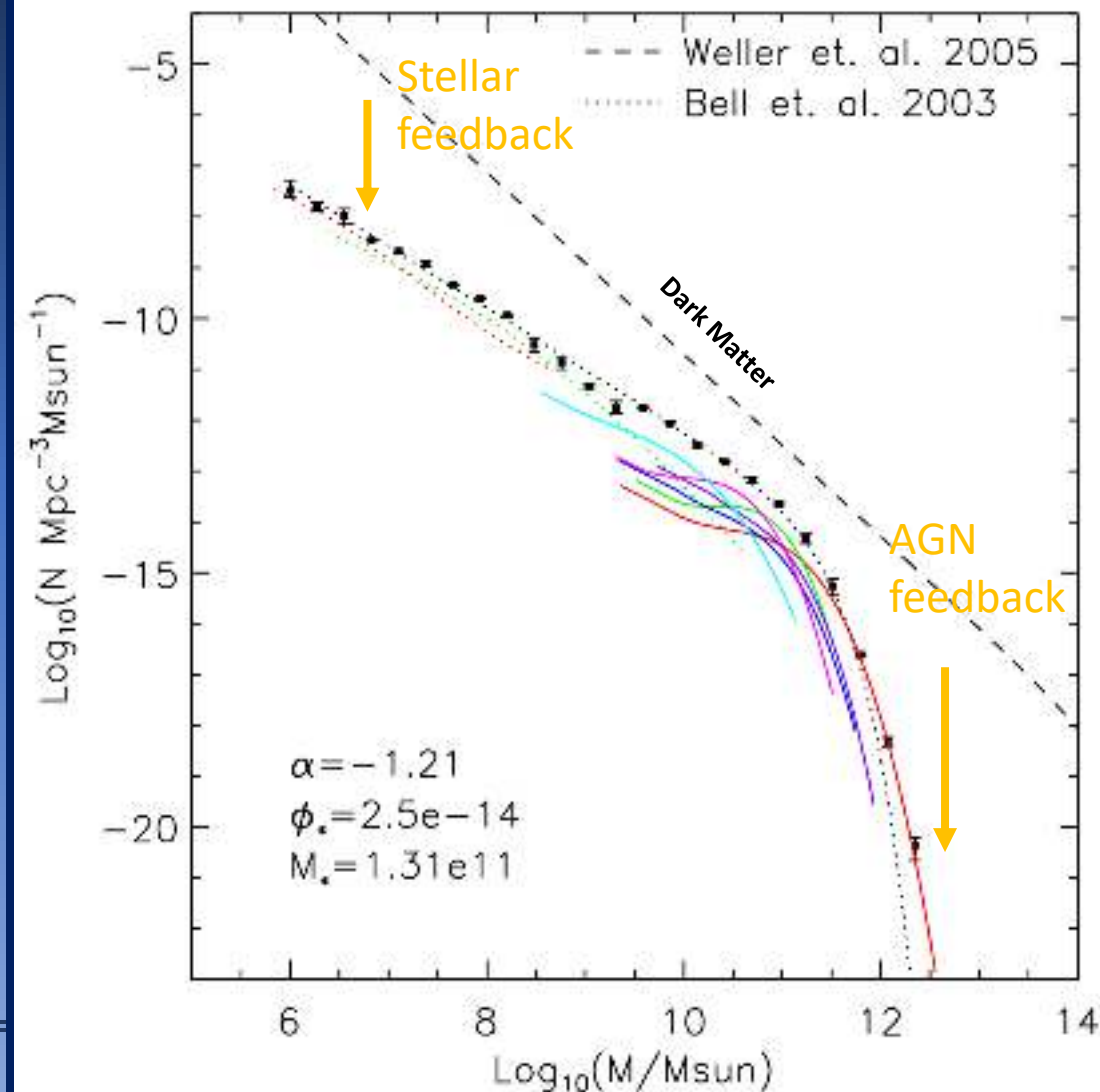




# Feedback

How feedback influences galaxies

... but it burns holes into disks

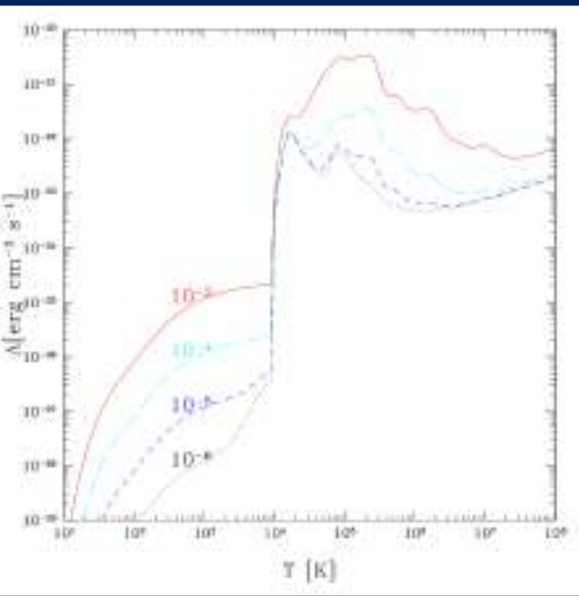


Read & Trentham 2005



# Summary: Including Physics

## Cooling



Maio et al.,  
2007

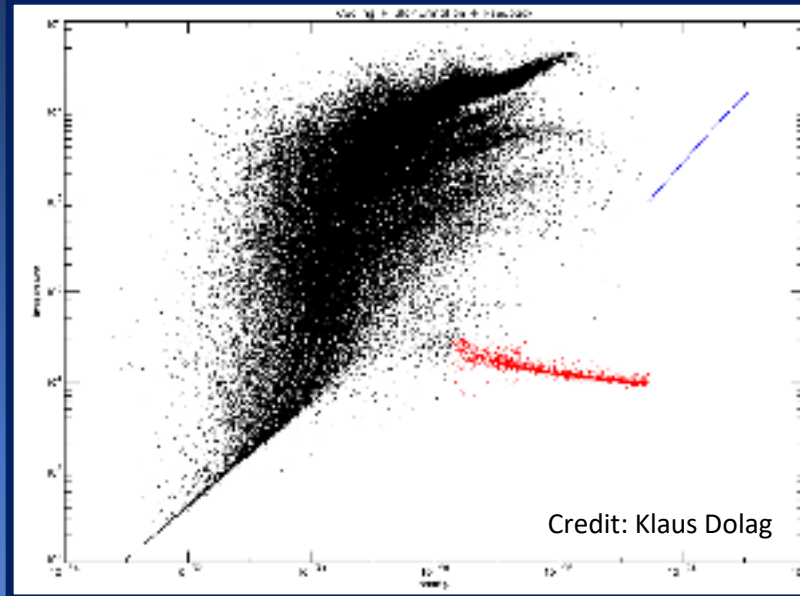
Basic Assumption:

- Optically thin
- Ionization equilibrium ( $H$ ,  $H^+$ ,  $He$ ,  $He^+$ ,  $He^{++}$ ,  $e^-$ )
- 2-body processes ( $\sim n^2$ )

$$\Lambda(T)/n^2$$

BUT: Cooling Catastrophe

## Star Formation



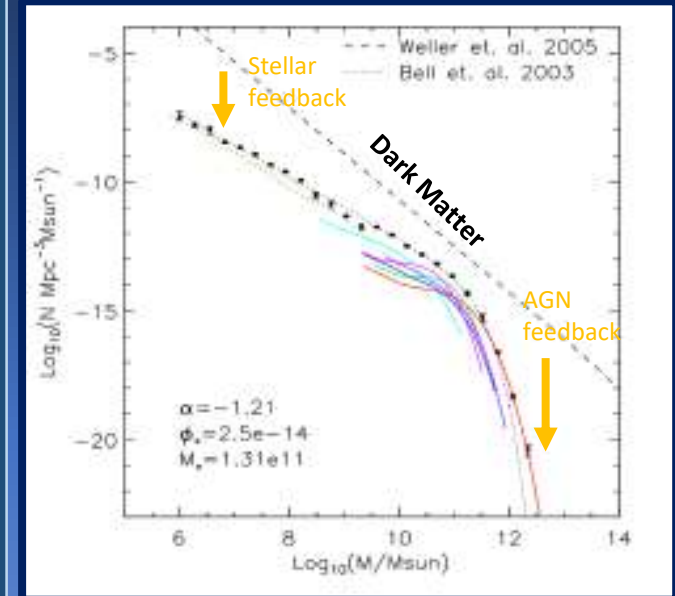
Credit: Klaus Dolag

Star formation subgrid model:

- Self-regulated star formation
- Set of differential equations needs to be solved.
- Produces reasonable galaxies at low  $z$

BUT: star formation rates at high  $z$  not captured

## Feedback



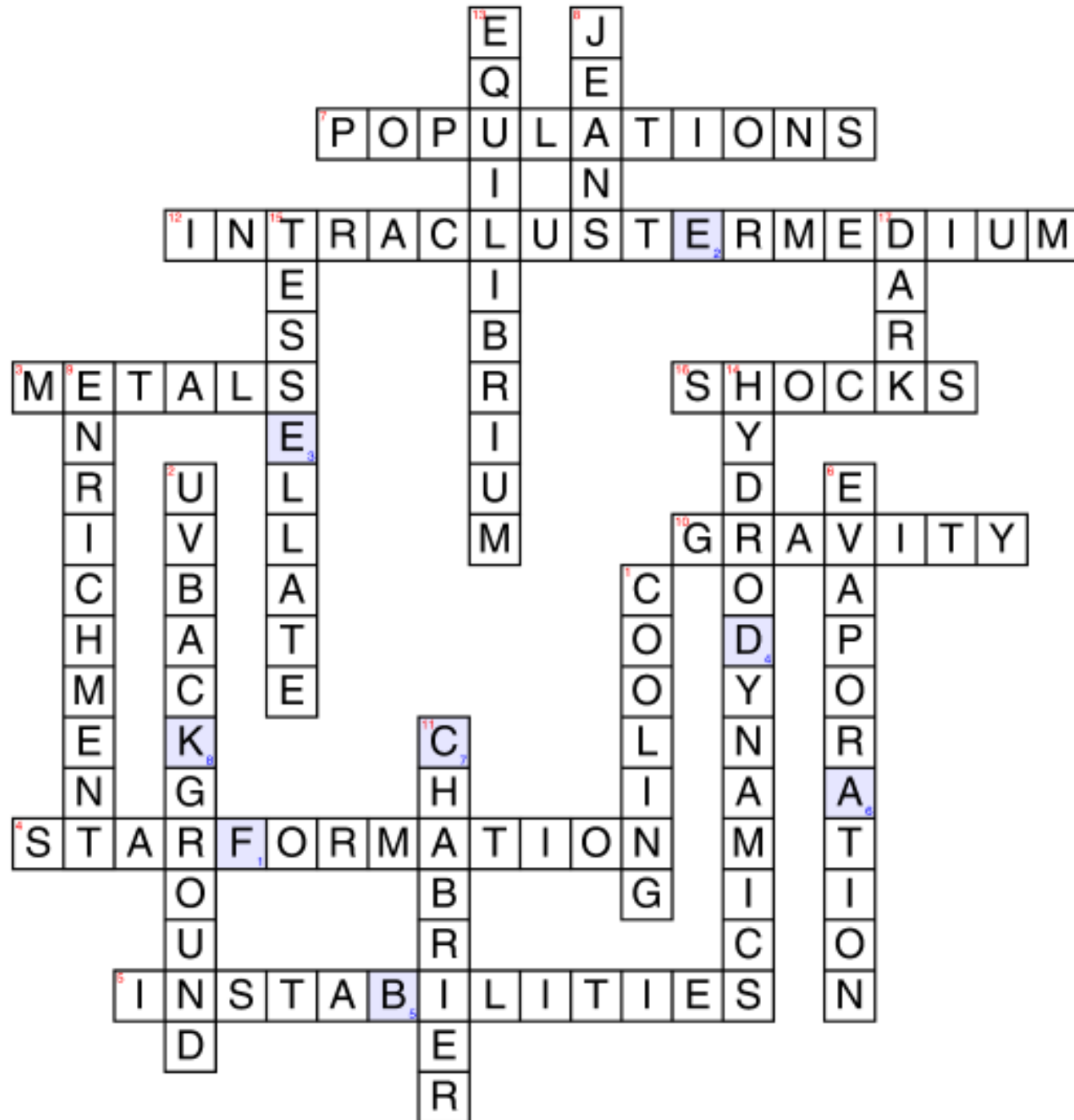
Read &  
Trentham  
2005

Feedback comes from two different sources:

- Massive Stars and Supernovae
- Supermassive Black Holes (AGN)

Stops the Overcooling Catastrophe

BUT: Burns holes into disks



- Necessary for gas to be able to form stars
- Drives ionization in the early universe
- Everything not hydrogen or helium
- Basic process that transforms gas
- Thermal, Kelvin–Helmholtz, Rayleigh–Taylor
- Radiation from (hot) stars causes cloud \_\_\_\_\_
- What star particles and countries have in common
- A criterion for collapse but also a fashion item
- Happens to gas through stellar death and at the stock market
- The force that dropped an apple onto Newton
- An IMF
- Diffuse component in a galaxy cluster
- Time-independent predictions are only possible for \_\_\_\_\_
- Basic physics that describes baryons
- To partition a plane into regions with certain properties
- Caused by high-Mach-number flows
- Most of the matter is like this

FEEDBACK